DEVELOPMENT OF AN R PACKAGE (aLBI) FOR ESTIMATING LIFE HISTORY PARAMETERS & STOCK STATUS FROM LENGTH FREQUENCY DATA

A Thesis Submitted for Partial Fulfillment of the Requirements for the Degree of Master of Science (M.S.) in Fisheries Management.

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Department of Fisheries
University of Chittagong
Chattogram - 4331, Bangladesh

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ABSTRACT

Stock assessment is a difficult task for sustainable fisheries management especially when population and fisheries data are limited. This study aims to develop an accessible and tested R package (aLBI) that applies length frequency and provides functions for estimating length-based indicators (LBIs) to measure stock health. The primary objective is to streamline the calculation and visualization of the estimated metrics for a clear understanding. The package includes three main functions, i.e., FishPar (for calculating key LBIs such as maximum length (L_{max}), asymptotic length (L_∞), length at sexual maturity (L_{mat}), optimum length (L_{opt}), $L_{opt\pm10}$ % ($L_{opt\_p}10$, $L_{opt\_m}10$), and percentages of fish at maturity (P_{mat}), optimum length (P_{opt}), and mega spawners (P_{mega}) with confidence intervals), FishSS (for evaluating stock biomass status), and LWR (for visualizing length-weight relationship). This study also evaluated the performance of the package by applying length frequency data of *Sperata aor* collected from Kaptai Lake. The findings with a calculated optimum bin size of three (cm) demonstrate its utility in producing acceptable stock assessments, with mean estimates for L_{max}, L_∞, L_{mat}, and L_{opt} at 78.01 cm, 82.12 cm, 43.73 cm, and 47.03 cm, respectively, as well as respective confidence intervals. Besides, percentages of sexually mature fish, optimum length, and mega spawners are 2.04 %, 2.38 %, and 0.35 %, against targets of 100 %, 100 %, and 20 % respectively. By integrating advanced statistical methods and user-friendly interfaces, aLBI enables effective monitoring of fish populations and informed decision-making. Tested on various bin size (1–5 cm) with the collected length frequency data, the package performed consistently. Future recommendations include expanding functionality to cover more species and integrating additional biological parameters for enhanced stock assessments. Continued collaboration with the fisheries science community will ensure aLBI remains a cutting-edge tool for sustainable fisheries practices.

Key words: R package development, stock assessment, *aLBI*, length-based indicators (*LBIs*), target biomass, limit biomass, Kaptai Lake.

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ACRONYMS AND ABBREVIATIONS

Abbreviation	Elaborated Meaning
aLBI	A Length-Based Indicators (LBIs) R Packages
L_{inf} (L_∞)	Asymptotic Length. The maximum length at which no growth occurs
L_{max}	Maximum Length of the Individuals
L_{mat}	Length at First Sexual Maturity
L_{opt}	Optimal Length. Length at which maximum yield can be found
L_{opt_p10}	Optimum Length plus 10% (L_{opt}+10%)
L_{opt_m10}	Optimum Length minus 10% (L_{opt}-10%)
P_{mat}	Percentage of Sexually Mature Fish
P_{opt}	Percentage of Optimal Length Fish
P_{mega}	Percentage of Mega Spawners
P_{obj}	Objective Percentage (Sum of Pmat, Popt, Pmega)
LM_ratio	Ratio of Length at Maturity to Optimal Length (Lmat ÷ Lopt)
LCI	Lower Confidence Intervals
UCI	Upper Confidence Intervals
TSB40	Target Spawning Biomass at 40% of the unfished biomass
LSB25	Limit Spawning Biomass at 25% of the unfished biomass
R	Programming Language for Statistical Computing
CRAN	Comprehensive R Archive Network
URL	Uniform Resource Locator
Git	A distributed version control system used for tracking changes in source code during software development.
GitHub	Web-based Hosting Service for Version Control using Git
ELEFAN	Electronic Length Frequency Analysis
Commit	Saving changes made to files in the local repository
Push	Uploading local commits to a remote repository (e.g., GitHub, GitLab etc.)
GPL-3	General Public License Version - 3
devtools	Development tools package for R
knitr	Package for dynamic report generation in R
markdown	Package for Converting R Markdown Documents
testthat	Unit testing framework for R
readxl	Package to read Excel files into R
doi	Digital Object Identifier
VignetteBuilder	Tool for building vignettes in R packages
UTF-8	8-bit Unicode Transformation Format
UCrt	Unicode Consortium Unicode Technical Report

CHAPTER 01 INTRODUCTION

INTRODUCTION

1.1 Background and Motivation

Effective fisheries management has emerged as one of the most pressing global environmental concerns due to the rapid and widespread increase in over-exploited fish stocks (Hilborn & Walters, 1992; FAO, 2020). The sustainability of marine ecosystems and the economic livelihoods they support are fundamentally dependent on the development and implementation of sound management policies that promote the recovery and long-term health of these vital resources. A central pillar of this effort is scientifically robust fish stock assessment, a discipline that provides reliable estimates of a fish population's status relative to critical biological reference points, such as biomass and fishing mortality. These estimates form the scientific foundation for informed policy-making and allow for the implementation of essential conservation measures, including fishing quotas, effort controls, and gear modifications (Pilling et al., 2008; Punt et al., 2016). By balancing resource exploitation with conservation, these measures are essential for reducing fish stock depletion rates and preventing the catastrophic collapse of fisheries (Punt et al., 2020a). Therefore, a thorough and up-to-date understanding of the population status is critical for effective management (Alam et al., 2021).

Globally, the fisheries sector plays a significant and multi-faceted role in food security, economic development, and employment. The Food and Agriculture Organization (FAO) reports that over three billion people rely on fish as a primary source of animal protein, and fisheries and aquaculture provide livelihoods for approximately 12% of the world's population (FAO, 2020). However, this vital sector is under immense pressure. The FAO estimates that over 30% of global fish stocks are overfished, a trend exacerbated by habitat destruction, climate change, and weak governance structures. This makes the development of effective management strategies a matter of urgent global concern. The fisheries sector is particularly crucial for Bangladesh, contributing over 3.57% to the nation's Gross Domestic Product (GDP) and supporting the livelihoods of millions of individuals (DoF, 2023). Despite its critical importance, Bangladesh faces significant challenges, including overfishing and the degradation of aquatic ecosystems (Shamsuzzaman et al., 2017), which mirrors the broader issues confronting fisheries in many developing regions.

Precise fish stock assessment is necessary to address these challenges. It allows for the determination of fishing quotas, the identification of overfished stocks, and the implementation of conservation measures. Without reliable stock assessments, fisheries management cannot effectively balance exploitation with conservation, leading to the depletion of fish stocks and the collapse of fisheries (Pilling et al., 2008).

Conventional methods for estimating fish numbers, such age-dependent treatments, typically call for extensive and reliable data collecting—which can be both expensive and time-consuming. These techniques depend on age estimate via otolith reading, a time-consuming and quite prone to major uncertainty process (Campana, 2001). By looking at the otoliths the bones in charge of hearing one can determine fish's age. This method breaks down, polishes, and under a microscope examines development rings. The activity is time consuming and demanding of significant knowledge; consequently, faults resulting from subjective interpretation are probably (Campana & Simon, 2001). Moreover, these strategies may not be helpful in underdeveloped nations since their availability of resources for extensive data collecting and analysis is limited (Pauly & Morgan, 1987).

Conversely, in cases of limited data access, length-based stock evaluation methods provide a good and typically more practical option (Pauly & Morgan, 1987). Using length frequency data, these techniques estimate significant population parameters like patterns of recruitment, growth rates, mortality rates due from natural causes and fishing, and so on (Sparre & Venema, 1998). One consistent approach to find length frequency information is to look at fish landings or market samples. These techniques are economically priced and readily accessible for many fishing businesses especially in developing nations (Gayanilo & Sparre et al., 2005). For fisheries management especially in areas with low financial and technological resources, the simplicity and cost-effectiveness of the length frequency data collecting method appeal (Hoggarth et al., 2006).

Based on length measurements, the *ELEFAN* (Electronic Length Frequency Analysis) method has been extensively applied in fisheries all around the world as it can rapidly assess the situation (Pauly & David, 1981). These methods estimate growth parameters by means of growth curves applied to length frequency data, therefore enabling deducing of other features of population dynamics. While catch curve analysis can give approximates of mortality rates (Beverton & Holt, 1957; Pauly, 1980), the von Bertalanffy Growth Function (*VBGF*) is usually used for growth rate modeling. Still, despite their great frequency, these methods usually rely on oversimplified presumptions—such as uniform rates of recruitment and death—which could not be true in every situation (Pauly, 1987; King, 2007).

Although numerous tools and approaches are available for assessing stock based on length, many existing R packages have limitations in terms of their adaptability, applicability, and possibility to add complex statistical models. Software applications such as "*TropFishR*", "*fishmethods*", "*LB-SPR*", and "*LBB*" for example, offer helpful features but might not completely meet the multiple needs of fisheries scientists and managers in different circumstances (Mildenberger et al., 2017; Nelson, 2017; Hordyk et al., 2015a; Prince et al., 2015; Hordyk et al., 2015b; Pauly & Froese, 2020). The difficulties highlight the need of developing a basic R package with uniform tools for managers and fisheries researchers that can close current gaps.

These constraints motivate programmers to create a fresh R tool for length-based stock research. Modern statistical methods including Bayesian approaches and machine learning algorithms, which can considerably increase the accuracy and dependability of stock assessments (Punt et al., 2020b) might not be able to be included into current tools or have the flexibility needed to fit different fisheries environment. Bayesian methods, for instance, allow one to add previous data and can more effectively control uncertainty, therefore providing a more comprehensive picture of market movements (Gelman & Shalizi, 2013). Furthermore, machine learning techniques help one to identify trends and anomalies in length frequency data, thereby checking the veracity of death and growth projections (Olden et al., 2008).

Furthermore, accessibility and user-friendliness are absolutely vital to guarantee that fisheries managers might optimize these tools without depending on major statistical programming expertise (Wickham, 2015). A well-designed R package with basic functions, extensive documentation, and interactive visualization components assisting to adopt length-based evaluation methodologies can benefit many diverse fisheries situations.

1.2 Problem Statement

Despite the critical need for sound scientific assessment, traditional, data-intensive stock assessment methods remain challenging to apply to most fish stocks, particularly in developing countries. These methods, such as age-structured models, require extensive and high-quality time-series data on catch, effort, and life history parameters, which are often unavailable or unreliable. For example, age estimation via otolith reading is a time-consuming, expensive, and technically demanding process that is prone to significant uncertainty due to subjective interpretation (Campana, 2001; Campana & Simon, 2001). The lack of institutional resources, infrastructure, and trained personnel further compounds these data limitations, hindering effective fisheries management in these regions (Hilborn and Walters, 1992; Scott et al., 2016).

In response to these challenges, a variety of alternative stock assessment methods have been developed for data-limited fisheries. Some of these methods require yearly landing data and life history parameters (Martell & Froese, 2013; Thorson & Cope, 2015), while others utilize more readily available data, such as length-frequency data (Hordyk et al., 2014; Nelson, 2017; Froese et al., 2018; Rudd & Thorson, 2018; Ault et al., 2019). While these methods have improved assessment capabilities, the software tools available to implement them still present significant limitations. Current R packages for length-based stock assessment, while useful, frequently lack the adaptability and user-friendliness needed to meet the diverse objectives of modern fisheries managers. Their general lack of customizable options for parameter estimation and model fitting reduces the relevance of these tools to specific fisheries or stock assessment scenarios (Nelson, 2019).

Moreover, many existing tools fail to incorporate the most recent advancements in computational methods and statistical modeling, which are crucial for improving the reliability and accuracy of stock assessments. For example, traditional methods may not fully account for the inherent variability and uncertainty in length-frequency data, leading to potentially biased or inaccurate estimates (Haddon, 2011; Hilborn et al., 2020). The demand for more sophisticated technologies, such as hierarchical Bayesian models and machine learning algorithms, is evident (Punt, et al., 2020b), yet these are largely absent from or poorly integrated into existing software solutions. Finally, a significant barrier to the widespread adoption of these tools is their steep learning curve, which often requires a high degree of statistical programming expertise (Wickham, 2015). This lack of intuitive interfaces and comprehensive user instructions hinders effective implementation by managers who need to make rapid, informed decisions in the field.

1.3 Rational and Justification

The rationale for this study is to directly address the identified gaps by providing a comprehensive tool for length-based fish stock assessment. The development of a sophisticated R package is supported by three key aspects: scientific advancement, user accessibility, and flexibility.

This study aims to fill a significant void in the current suite of available tools by incorporating cutting-edge advancements in statistical and computational methods into a single, cohesive software package. While existing packages are valuable, they often fail to leverage the full power of modern methodologies. The "aLBI" package is designed to provide robust parameter estimation that accounts for the inherent complexity and uncertainty within fish population data. It will go beyond simple point estimates by providing outputs that are both scientifically rigorous and readily interpretable by managers. The R package provides a powerful and flexible tool for computing biological parameters, assessing stock status, and analyzing length-frequency data, thereby supporting informed judgments in fisheries management. This approach enhances the accuracy and reliability of stock assessments, moving the field forward, particularly for data-limited fisheries.

1.4 Aims and Objectives

This study's primary aim is to develop and validate a scientifically sound, user-friendly, and robust R package, "aLBI," for length-based fish stock assessment in data-limited contexts.

The specific objectives of this study are:

1. To develop a flexible R package with core functions that estimate life history parameters Froese's length-based indicators (LBIs from length-frequency data.
2. To create comprehensive documentation and user support, to enhance the package's accessibility and usability for a broad audience.
3. To validate the functionality and accuracy of the "aLBI" package by applying it to a real-world dataset of *S. aor* from Kaptai Lake.

1.5 Thesis Structure

This thesis is organized into five chapters. Chapter 1 provides a comprehensive introduction to the global context of fisheries management, outlines the specific problem of data-limited assessment, and presents the rationale and objectives for the development of the "aLBI" R package. Chapter 2 will review the relevant literature on both traditional and length-based stock assessment methods, with a specific focus on Froese's LBIs and the Cope and Punt (2009) decision-tree approach. Chapter 3 will detail the materials and methods used for the package's development and validation, including a description of the data collection and the analytical procedures. Chapter 4 will present and discuss the results of the case study on *S. aor* from Kaptai Lake. Chapter 5 will summarize the study's conclusions, highlight its key contributions to the field, and provide recommendations for future research. Chapter 6 will contain the recommendations for management of sustainable fisheries in Kaptai Lake. Chapter 7 will document all the references used in this study. Finally, Chapter 8 will detail about the vignettes of aLBI R package.

CHAPTER 02 LITERATURE REVIEW

2 LITERATURE REVIEW

2.1 Historical Context and Importance of Stock Assessment

Fish stock assessment is a foundational pillar of modern fisheries management, providing the necessary scientific framework to evaluate the health and sustainability of fish populations. Its historical trajectory has been driven by the increasing need to manage human exploitation of marine resources. Early fisheries science, formalized by pioneers like Beverton and Holt (1957) and Ricker (1954), focused on understanding population dynamics through age-structured models. These foundational approaches, which quantified the relationship between stock biomass and recruitment, were instrumental in the development of yield-per-recruit and spawner-per-recruit models, still in use today. However, these methods, while powerful, relied on extensive, high-quality data on catch, effort, and age composition, which were often difficult and expensive to obtain.

This limitation led to the increasing significance of length-based stock assessment, particularly for fisheries in developing regions. Length frequency data are often simpler and less expensive to collect than age data, which requires labor-intensive methods like otolith reading (Quinn & Deriso, 1999; Haddon, 2011). The work of Pauly (1980) was pivotal in this area, as he developed a widely accepted method to relate natural mortality, growth criteria, and environmental temperature. Similarly, the introduction of the ELEFAN (Electronic Length Frequency Analysis) method by Pauly and David (1981) demonstrated that growth parameters could be reliably estimated from length frequency data, a method that has since been successfully applied in numerous fisheries globally (Gayanilo & Pauly, 1997). The Food and Agriculture Organization (FAO) has actively promoted these length-based methods for their simplicity and cost-effectiveness, especially in tropical fisheries where age identification can be particularly challenging (Pauly & Morgan, 1987; FAO, 2001).

Dichmont et al. (2021a) claim that stock evaluations are essential for fisheries management and have transitioned toward more sophisticated, flexible, and documented software tools to improve efficiency, consistency, and dependability. A new website was created to compile and showcase available stock assessment tools, providing a centralized platform that regularly highlights their characteristics and allows for user feedback. This initiative helps to foster the creation of next-generation stock assessment packages. Furthermore, in contrast to biomass-aggregated surplus production models (SPMs), age-structured production models (ASPMs) are increasingly recommended for tracking cohort propagation and selective fishing effects. JABVA-Select, for example, is an extension of the

JABVA software that incorporates life history data and fishing selectivity to overcome common SPM drawbacks, providing accurate estimates of spawning biomass and stock status uncertainty (Ault et al., 2019).

Fish stock assessment is absolutely crucial for fisheries management since it helps to assess degree of exploitation, sustainability, and trend of fish population. Among the classic stock evaluation systems are age-based, biomass-based, and length-based ones. Length-based stock assessment has been increasingly significant due to its relative simplicity and the availability of length frequency data often simpler and less expensive than age data because of which it has become more relevant (Quinn & Deriso, 1999; Haddon, 2011). Particularly useful in data-limited environments, length-based techniques provide vital insights when other methods would be unworkable (Hilborn & Walters, 1992). Understanding recruitment and death trends requires a high-resolution view of population structure, which studies has also shown may be given using length frequency data (King, 2007).

Dichmont et al. (2021b) claim that stock evaluations are essential for fisheries management; hence, assessments applying population dynamics models are in increasing demand. Although he is now employing flexible, documented, and tested software tools, which improve efficiency, consistency, and dependability, hetaeristically dependent on unique approaches. A new website has been created to compile accessible stock assessment tools, therefore offering a single platform that regularly highlights their characteristics. By allowing user comments and stressing present state-of-the-art tools, this project helps to encourage the creation of next-generation stock assessment packages.

More studies emphasize how important length-based assessments are for the sustainable control of fish populations. Pauly (1980), for instance, developed a generally accepted method to relate natural death, growth criteria, and environmental temperature with one another. Moreover, the foundation for still-in use yield-per-recruit models today was laid by Beverton and Holt (1957). Length-based techniques also show value for tropical fisheries, where age identification might be difficult (Pauly & Morgan, 1987).

The application of length-based assessment methods in several fisheries is extensively documented. Pauly and David (1981) for example presented the ELEFAN method, which uses length frequency data to estimate growth parameters. This method has been used successfully in several fisheries all across the world (Gayanilo & Pauly, 1997). Moreover, the FAO has promoted length-based methods for usage in underdeveloped countries (FAO, 2001) because of their simplicity and economy.

Apart from useful applications, theoretical developments have been rather important. Emphasizing the use of length-based techniques, Sparre and Venema (1998) produced a thorough guide on tropical fish population evaluation. Moreover, Gulland and Rosenberg (1992) talked about the need of combining length-based approaches with other stock assessment tools to enhance management decisions.

For tracking cohort propagation and selective fishing effects, however, age-structured production models (ASPMs) are recommended over biomass aggregated surplus production models (SPMs). Incorporating life history data and fishing selectivity helps JABVA-Select, an extension of the JABVA software, overcome typical SPM drawbacks. Applied to South African Silver kob, JABVA-Select exceeded the Pella-Tomlin son model and matched ASPMs in accuracy, therefore providing accurate estimates of spawning biomass and stock status uncertainty. It is a strong tool for data-moderate stock evaluations.

2.2 Fisheries Management and Stock Assessment

The primary goal of fisheries management is to ensure the long-term sustainability of fish stocks, marine ecosystems, and the human communities that depend on them. This is achieved through the application of various management strategies, such as setting total allowable catches (TACs), implementing fishing quotas, controlling fishing effort, and protecting critical habitats (Pilling et al., 2008). To be effective, these strategies must be grounded in robust scientific evidence. This is where stock assessment plays a critical, symbiotic role. By estimating key population parameters—such as growth, mortality, and recruitment—and evaluating the current status of the stock relative to biological reference points (e.g., maximum sustainable yield, MSY), stock assessment provides the scientific basis for management decisions.

The concept of biological reference points is fundamental to this process. Target reference points (TRPs) define a desirable state for a stock (e.g., a biomass level that produces MSY), while limit reference points (LRPs) define an undesirable state that should be avoided (e.g., a biomass level below which recruitment is impaired). The precautionary approach to fisheries management, endorsed globally by the FAO, emphasizes the need for action even in the face of scientific uncertainty to prevent irreversible damage to a stock. This requires tools that can provide reliable estimates even with limited data, which has become a driving force behind the development of novel assessment methods.

2.3 Data-Limited Stock Assessment Methods

For many of the world's fisheries, particularly small-scale and artisanal fisheries, the data required for traditional, data-intensive stock assessment methods are simply not available (Hordyk et al., 2014; Froese et al., 2018). In response to this pervasive problem, a new class of data-limited stock assessment methods has emerged. These methods provide a good and typically more practical option for managers in data-poor situations (Pauly & Morgan, 1987).

These approaches are diverse, ranging from methods that require minimal yearly landing data and life history parameters to those that rely solely on length-frequency data (Martell & Froese, 2013; Thorson & Cope, 2015). Length-based methods have gained particular prominence because the data are often more straightforward and cost-effective to collect, and the resultant information tends to be more reliable and less susceptible to the biases associated with other data types (Chong et al., 2020a; Pilling et al., 2008). By analyzing length frequency distributions, these techniques can estimate significant population parameters, including recruitment patterns, growth rates, and mortality rates (Sparre & Venema, 1998).

2.4 Length-based Stock Assessment Techniques

As stated by (Cope & Punt, 2009), current fisheries management strategies usually reevaluate stock status, a challenging chore given population and fisheries data are inadequate. Current fisheries management strategies measure population relative to exploitation by means of catch length compositions (Pmat, Popt, Pmega), so guiding stock status. Together referred to as Px, these indicators seek to prevent overfishing but have no quantifiable relationship to stock state or sustainable catches. Using simulations, the paper investigates Px's relationship with fishing mortality and spawning biomass, so stressing the impact of fisheries selectivity, life cycle features, and recruitment compensation. Combining Pmat, Popt, and Pmega, a new metric Pobj generates stock state indicators and harvest control rules, therefore supporting data-limited fisheries management.

Examining fish population size helps length-based methods assess fish availability. Length frequency data allows techniques including length cohort analysis (LCA) and the Beverton-Holt yield-per recruit model estimate factors including growth rates, mortality rates, and recruitment patterns (Sparre & Venema, 1998). Another popular method including tools for length-based assessment including ELEFAN and other growth models is the FiSAT II (FAO-ICLARM Stock Assessment Tools). For species for which age assessment is difficult or unreliable, these methods are very useful. Applied in data-limited fisheries, the length-based approach allows rapid evaluations (Pauly & Morgan, 1987). In multi-species fisheries, where they can provide perceptive study of the dynamics of numerous species inside the same ecosystem, length-based methods have also shown their strength (Hoggarth et al., 2006).

Recently on the scene are advanced statistical techniques and models meant to increase the accuracy of length-based tests. Using maxi mum likelihood estimation and Bayesian techniques has helped the precision of parameter estimates (Lai & Gallucci, 1988; Millar & Fryer, 1999) to get better. Furthermore, developed to provide more full assessments are integrated assessment models combining length-based data with other types of data (e.g., catch and effort data).

Case studies demonstrating the successful application of these techniques abound. For instance, LCA was applied by Sparre and Venema (1998) to assess several tropical fish species' populations, therefore providing useful management guidance. Using yield-per--recruit models, Beverton and Holt (1957) also helped North Sea fisheries to be managed. Moreover, well-documented are the application of length-based techniques in data-limited fisheries, which underlines their importance in management of sustainable fisheries (Pauly, 1987; King, 2007).

The R package TropFishR developed by (Mildenberger et al., 2017) offers tools for single-species stock assessment using length-frequency data, tailored for data-limited fisheries. It estimates growth and mortality parameters, explores fisheries' technical aspects, conducts virtual population analysis (VPA), and assesses stock status with yield prediction and production models. This paper introduces TropFishR, showcasing its core methods and modernizing traditional assessment methods with advanced statistical approaches.

Alam et al. (2022) assessed the stock status of Bombay duck fishery in Bangladesh from length-based. This study found the stock as crucial for the local economy, is overexploited with a spawning biomass (SPR = 8%) below sustainable levels due to juvenile-oriented fishing practices. This study recommends implementing a 5 cm mesh size regulation for set bag nets and halving the number of SBNs to protect immature fish and reduce fishing pressure, ensuring sustainable management of the fishery.

Barua et al. (2022) evaluated the stock status of the black pomfret (*Parastromateus niger*) using TropFishR (Mildenberger et al., 2024), LB-SPR (Hordyk et al., 2015), and Sustainability indicators (LBIs) of Froese (2004) revealing overexploitation and low spawning biomass (SPR = 13%). Key findings include an allometric growth pattern and high fishing mortality (F = 0.91 year−1). Recommendations include enforcing a minimum mesh size to avoid catching immature fish, targeting catch lengths between 29-35 cm, and reducing fishing pressure by one-third to enhance sustainability.

Froese et al. (2018) developed the Length-Based Bayesian (LBB) method analyzes length frequency data to estimate key parameters like asymptotic length, relative natural mortality, and fishing mortality. It approximates current biomass relative to unexploited biomass and provides a new stock health indicator. LBB's estimates are consistent with simulations and independent assessments but require representative length frequency data for accuracy.

Three fisheries indicators are proposed by (Froese, 2004): percentage of mature fish in catch (target 100%), percentage of fish at optimum length (target 100%), and percentage of mega-spawners (target 0%, 30-40% for reasonable stock structure). Applied to stocks of Gadus morhua, Sardinella aurita, and Epinephelus aeneus, these indicators engage diverse stakeholders in managing and reversing "convenience overfishing."

According to Canales et al. (2021), for many medium and small-scale fisheries lacking catch data and scientific observer programs, length-frequency data offers a cost-effective alternative for stock assessment. Therefore, a Length-Based Pseudo-cohort Analysis (LBPA) model was developed to estimate parameters using multiple length frequencies, reducing the impact of equilibrium assumptions. Simulations showed LBPA outperformed the Length-Based Spawning Potential Ratio (LBSPR), especially with high exploitation rates, offering more accurate and less biased estimates. This study provides guidelines for using length-based pseudo-cohort models in data-poor fisheries.

Xu et al. (2023) assessed the Mackerel Scad (*Decapterus macarellus*) in the South China Sea by means of length-based Bayesian biomass estimate and length-based spawning potential ratio techniques. Results reveal overfishing; relative stock size falls from 1.3 to 0.7 and spawning potential ratio falls from 13% to 12%, below the 20% reference point. Among the suggestions include increasing worldwide stock conservation cooperation and lowering the minimum mesh size in fishing nets.

Dai et al. (2021a) look at the most recent methodological developments in length-based approaches aimed to support data-limited fisheries management. The effect of systematic observation errors on the length-based spawning potential ratio (LBSPR) estimation was explored in this work under simulation-estimation analysis. Emphasizing the need of species-specific assessments including such biases, results revealed that LBSPR performance is somewhat sensitive to observation errors, especially in cases of small fish overrepresentation.

Heery & Berkson (2009a) delighted in length frequency data since systematic sampling errors can skew age-structured models used in fisheries stock estimates. Using simulations, their study investigated the effects of such biases and found that data skewed towards larger fish resulted in too optimistic stock status estimations, therefore risking poor management activities; data slanted towards smaller fish could cause too strict measures. The results underline the need of considering possible biases in length frequency data while choosing modeling techniques for stock assessments.

Under several circumstances, Chong et al. (2020b) simulated estimated four length-based stock evaluation techniques (TB, LBSPR, LIME, and LBRA). While LBRA was highly biassed yet preventative, LIME was appropriate for longer time-series assessments; TB and LBSPR were the most consistent and accurate. Emphasizing the need of testing these strategies under several conditions to understand their strengths and limits, all approaches struggled with short-lived species and were less accurate under severe overexploitation or significant recruitment error.

2.5 Statistical and Mathematical Models in Length-based Stock Assessment

The foundation of length-based stock assessment is built on robust mathematical and statistical models. Key among these is the von Bertalanffy Growth Function (VBGF; von Bertalanffy, 1938). The parameters L_∞ (asymptotic length) and K (growth coefficient) are central to length-based models. Length-converted catch curves, based on the work of Beverton and Holt (1957) and Pauly (1980), are used to estimate total mortality (Z) from length frequency data, which can then be partitioned into fishing mortality (F) and natural mortality (M). These models are often integrated into software like FiSAT II and R packages to provide a suite of analytical tools.

Recent developments have moved toward more complex and flexible models. Bayesian statistical methods have gained prominence for their ability to incorporate prior information, quantify uncertainty, and provide more comprehensive estimates of population dynamics (Gelman et al., 2013; Punt et al., 2020a). Stochastic models and simulation approaches have also been developed to address the inherent variability in length frequency data, offering more accurate and robust estimates (Haddon, 2011; Hilborn et al., 2020). Furthermore, the integration of environmental factors into length-based models has been a key area of development, as factors such as temperature, food availability, and habitat conditions significantly influence growth and mortality rates (Myers & Worm, 2003a; Brander, 2007). The use of machine learning (ML) models, as proposed by Ludtke & Pierce (2023a), has also shown promise in improving the accuracy of fish stock assessments by correcting for post-hoc errors and enhancing forecast accuracy.

Strong mathematical and statistical models provide the foundation of length-based stock assessment. While length-converted catch curves help estimate mortality rates (Beverton & Holt, 1957; Pauly, 1980), models such as the von Bertalanffy Growth Function (VBGF) describe growth patterns. Often utilized is the ELEFAN (Electronic Length Frequency Analysis) approach, which fits the VBGGF to length frequency data (Pauly & David, 1981). Recent developments include Bayesian methods and machine learning techniques improving model accuracy and adaptation to several fish species and environmental variables (Punt et al., 2020a). Furthermore, designed to solve the uncertainty and variability inherent in length frequency data are stochastic models and simulation approaches, thereby offering more accurate estimates. (Haddon,2011; Hilborn et al., 2020)

Significant contributions have been made over years to help statistical models for length-based assessments evolve. For example, Ricker (1975) developed models for stock-recruitment interactions, which are basic for understanding of population dynamics. Moreover, well studied has been the fitting of growth models by non-linear regression techniques (Fabens, 1965; Schnute, 1981).

However, under development has been the integration of environmental factors into length-based models. Studies of components including temperature, food availability, and habitat conditions have shown how much they can affect rates of growth and mortality (Myers & Worm, 2003b; Planque et al., 2010). Including these elements into length-based models can thus improve their accuracy and prediction capability (Brander, 2007).

More complex models can now be developed because to recent improvements in computational techniques. Hierarchical Bayesian models, for instance, enable the integration of several sources of uncertainty and variability, hence strengthening estimates (Gelman et al., 2013). Applied to length-frequency data, machine learning techniques have also presented fresh opportunities to enhance stock ratings (Olden et al., 2008).

Van Nes et al. (2002) look at how difficult it is to untangle the processes influencing multi-species fish community dynamics given complicated relationships and size variances within populations. Limited traditional aggregated models lead to the introduction of individual-based models such as PISCATOR, which deals with multi-species populations. Using a step-by-step calibration and Monte Carlo sensitivity analysis, PISCATOR manages complexity shown by a Frisian Lakes fisheries experiment by means of Attributed to community feedback, environmental variability, and a strong year-class development resulting from consecutive warm years, high bream catches were seen to have no effect on the population. This reasonable model helps one to grasp multi-year cycles and ecological processes.

Dai et al. (2021b)evaluated the performance of the length-based spawning potential ratio (LBSPR) method under various conditions, focusing on the impact of systematic observation errors, life history types, and selectivity. The findings revealed that LBSPR estimation is sensitive to these factors, especially when small fish are disproportionately represented, causing significant deviations in stock status estimates. The study highlights the need for careful spices-specific evaluation of length-based stock assessments when observation errors are present.

2.6 Length-Based Indicators (LBIs)

Froese (2004) introduced Length-Based Indicators (LBIs)—P_{mat} (percentage of mature fish), P_{opt} (percentage at optimum length), and P_{mega} (percentage of mega-spawners)—as practical tools to assess stock health. These metrics aim for targets of 100% for Pmat and Popt, and 30–40% for Pmega, guiding management to reverse overfishing. Case studies on *Gadus morhua*, *Sardinella aurita*, and Epinephelus aeneus demonstrated their engagement with diverse stakeholders (Froese, 2004). Cope and Punt (2009) combined these into a composite metric, Pobj, validated through simulations, though they lack direct ties to sustainable catches. LBIs' simplicity suits data-limited settings, but accuracy depends on representative length data (Froese et al., 2018). Their application in Bangladesh, as seen in Alam et al. (2021) for *Tenualosa ilisha*, highlights overexploitation issues, reinforcing their management relevance.

2.7 Existing R Packages and Software for Fish Stock Assessment

Fish stock studies can be conducted in several ways these days utilizing R packages such as "TropFishR" (Mildenberger et al., 2024) and "fishmethods" (Nelson, 2023). These initiatives project mortality, do cohort analysis, and provide us the tools required to create reasonable development models. For instance, "TropFishR" combines many length-based evaluation techniques with the ELEFAN approach. Furthermore, difficult to change models and requiring users to be well-versed in statistical techniques are these tools (Mildenberger et al., 2017). Useful programs available in both data-poor and data-rich environments are FiSAT II and Stock Synthesis (Methot & Wetzel, 2013; Gayanilo et al., 2005). International fisheries surveys extensively make use of them.

The development of these technologies has been driven by the need for cheaply cost, user-friendly tools for managers and fisheries scientists. FiSAT II, for example, has been especially used in undeveloped countries (Gayanilo & Pauly, 1997) because of its simplicity and efficiency in managing length frequency data. Comparably, Stock Synthesis is a flexible tool that may mix several types of data and provide complete evaluations for demanding fisheries (Methot, 2000). Methot and Wetzel, 2013)

R packages have changed lately with an aim toward more usability and utility. For fisheries study, for example, the "fishmethods" package provides a wide range of tools including methodologies for growth, mortality, and recruitment estimate (Nelson, 2019). 'TropFishR' offers a broad spectrum of tools that would be valuable for academics working in tropical fisheries (Mildenberger et al., 2017).

Another aim has been on including cutting-edge statistical methods onto these instruments. For "TropFishR," for example, the application of maximum likelihood estimation and Bayesian techniques enables more precise parameter estimate (Taylor & Mildenberger, 2017). Furthermore, including simulation tools into Stock Synthesis offers a way to assess the effects of several management approaches (Methot & Wetzel, 2013).

Developed by Hordyk et al. (2015), the length-based model (LB-SPR) calculates spawn potential ratio (SPR) using catch size composition, hence benefiting data-limited fisheries management. With appropriate input settings, simulations revealed the model's correctness with regard to non-equilibrium dynamics and sensitivity to recruitment variability. Although it depends on exact estimations of M/k, L_∞, and CVL_∞, empirical experiments verified its potential and meticulous validation of assumptions and biological factors.

Fish stocks interact through predation and resource competition, but management usually takes place apart from this. Acknowledging the need of multispecies interactions, LeMaRns, a new R-package by Spence et al., (2020a) offers a transparent, length-structured fish community model supporting mixed-fishery interactions and multispecies reference points. The speed, extensive documentation, open-source character of this program improves its accessibility for fisheries management.

Zhu et al. (2023) assessed the stock status of *Trichiurus lepturus* by using length-based Bayesian biomass estimation (LBB) from the East China Sea (2016-2020), the study found high exploitation rates and low biomass, indicating overfishing. Despite a 29% decrease in fishing vessels, high fishing pressure persists, suggesting the need to reduce fishing intensity, enforce size-at-first-capture regulations, increase mesh size, and reduce juvenile fish catch to recover the stock. The findings offer valuable insights for managing *T. lepturus* under data-limited conditions.

2.8 Data Requirements and Sources

The quality and reliability of length-based stock assessments are critically dependent on the quality of the underlying data. High-quality length frequency data, typically obtained via commercial catch records or fisheries-independent surveys, are the foundation of these assessments (Haddon, 2011). Key data requirements include accurate length measurements, representative sampling across seasons and habitats, and sufficient sample sizes to ensure good estimates (Sparre & Venema, 1998).

Common sources of length frequency data include fisheries management agencies, research facilities, and monitoring projects. However, challenges exist in guaranteeing adequate temporal and spatial coverage, addressing sample biases, and standardizing data collection (Jennings et al., 2001). For instance, Heery & Berkson (2009b) demonstrated that systematic sampling errors that skew data toward either larger or smaller fish can lead to biased stock status estimations, which could result in either overly optimistic or overly strict management measures. Creative data collection approaches, such as the use of electronic monitoring systems and citizen science projects, have been explored to improve data quality (Helmond et al., 2020). The challenges are particularly acute in inland fisheries, which are often highly diverse and seasonal, and where traditional data collection methods are often insufficient (Fitzgerald et al., 2018a; Hommik et al., 2020a).

Usually obtained via commercial catch records or fisheries-independent surveys, high-quality length frequency data provides the basis of length-based stock assessment (Haddon, 2011). Key data needs are accurate length measurements, representative sampling over seasons and habitats, and sufficient sample sizes to assure good estimates (Sparre & Venema, 1998). Common sources of length frequency data are fisheries management agencies, research facilities, and monitoring projects. Among the challenges include guaranteeing temporal and spatial coverage, addressing sample biases, and data standardization (Jennings et al., 2001). Furthermore, shown is the improvement in the quality and relevance of the acquired knowledge by merging scientific data with traditional ecological knowledge (Johannes et al., 2000).

For length-based assessments, several studies have underlined the need of precise and representative data. Allen and Hightower (2010), for instance, examined how sampling biases affected stock assessment findings and suggested strategies to minimize these biases. Likewise, Walters and Martell (2004) underlined the importance of thorough data collecting systems to guarantee the validity of length-based tests.

It is well known that fishery-independent surveys help to produce excellent data. Research organizations and management companies run these polls intended to gather objective, representative data on fish numbers (Pikitch et al., 2004). Though important, commercial catch records sometimes need meticulous validation and correction for biases linked to fishing methods and reporting (Branch et al., 2010).

Furthermore, investigated are creative approaches of data collecting to raise the quality of length frequency data. For fishing vessels, for example, the application of electronic monitoring systems has been demonstrated to yield accurate and real-time data on catch composition and size distribution (Helmond et al., 2020). Citizen research projects have also been rather successful in gathering vast amounts of length frequency data, especially in leisure fish parks (Bonney et al., 2009).

Fitzgerald et al. (2018b) stated inland fisheries as more diverse and highly seasonal, often lack sufficient data for traditional assessment methods. This study demonstrates the application of three data-poor assessment tools from marine systems to inland fisheries, using archived data to evaluate the ecological status of a brown trout stock in an Irish lake. The broader aim is to bridge marine and inland fisheries science and highlight accessible assessment methods for diverse inland fisheries management.

Hommik et al. (2020b) reported inland fisheries are often lack of monitoring and quantitative assessment, but the Length-Based Spawning Potential Ratio (LB-SPR) model, requiring only fish length data and life history parameters, shows potential. This study extended LB-SPR to accommodate dome-shaped selection curves typical of gillnets and hooks and applied it to brown trout in Irish lakes. The extended model produced reliable stock status estimates despite small sample sizes, demonstrating its applicability for data-limited inland fish stocks.

2.9 Developing R Packages for Stock Assessment

The development of R packages for fisheries science requires adhering to best practices for software development, including a modular architecture, comprehensive documentation, and user-friendly interfaces (Wickham, 2015). A primary objective is to create tools that are not only powerful but also accessible. This includes providing thorough examples, clear tutorials, and maintaining computational performance (Xie, 2018). Open-source concepts are particularly important, as they can help to increase the functionality of the package and build a community of contributors capable of managing the ever-shifting needs of fisheries research (Boehmke & Greenwell, 2020). For example, the s3sim package enables flexible and repeatable simulation testing of stock assessment models (Anderson et al., 2014), and the r4ss package enhances the efficiency of the entire assessment process by providing a wide range of functions for the Stock Synthesis modeling framework (Taylor et al., 2022).

While developing R packages for fisheries science, the best standards for software development modular architecture, extensive documentation, and user-friendly interfaces must be followed (Wickham, 2015). Xie (2018) notes that one of the primary issues is making sure the program offers thorough examples and tutorials while with maintaining computational performance. Working with fishery scientists, software developers, and statisticians will help to produce goods that satisfy a spectrum of consumer needs (Chambers, 2008). Open-source concepts can also help to increase the functionality of the package and build a community of contributors able of managing the often-shifting needs of fisheries research (Boehmke & Greenwell, 2020).

Development of R packages in fisheries science usually starts with the awareness of the particular requirements and difficulties faced by managers and researchers. For instance, "Trop Fish R" was created to fill in the void of thorough tools for tropical fisheries (Mildenberger et al., 2017). In the same vein, "fishmethods" were developed to give fisheries researchers a broad spectrum of analytical tools (Nelson, 2019).

Method evaluation of fisheries stock assessment depends on simulation testing. Using the extensively used Stock Synthesis (SS3) architecture, the R package s3sim created by Anderson et al. (2014) enables flexible, repeatable, and quick end-to- end simulation testing. It produces an underlying 'truth,' samples from it, runs estimating models, and synthesizes results to allow the investigation of structural variations, model misspecification, and data relevance in fisheries assessments.

Ludtke & Pierce (2023b) investigated how machine learning (ML) models might improve the accuracy of evaluations of fish stock. Proposed is a hybrid model integrating gradient boosted trees with standard statistical stock assessment methods, using initial estimations and using ML for post-hoc corrections. Five distinct stock experiments reveal statistically significant increases in the recruiting and spawning stock biomass forecast accuracy.

In software development, best practices stress the importance of modular design, which provides simplicity of maintenance and flexibility. Wickham (2015), for instance, guides on the use of functions and classes to capture certain operations, therefore boosting the readability and modification simplicity of the code. Moreover, very important is comprehensive documentation since it provides users with the knowledge needed to understand and maximize the package (Xie et al., 2018).

The success of R packages relies on end user and developer interaction. Interacting with the fisheries science community guarantees that the package meets user needs by way of seminars, user feedback sessions, and cooperation projects (Chambers, 2008). Moreover, an open-source strategy can support cooperation and innovation as it lets users help the package to be developed and improved (Boehmke & Greenwell, 2020).

Packages considerably expand R's value and help to improve inadequate data science solution approaches. R is a preferred tool environment for statistical computation and data analytics more and more according to (Wendt & Anderson, 2022) among data scientists and practitioners.

According to Taylor et al. (2021) stock assessment analysts are increasingly using diverse models while maintaining high standards for consistency, documentation, and transparency within tight timelines. The r4ss package by Taylor et al. (2022) which supports the Stock Synthesis modeling framework, has significantly enhanced efficiency across the assessment process. Originating as a tool for simple model diagnostics 15 years ago, r4ss now encompasses a wide range of R functions for various assessment tasks. This overview highlights r4ss features, its practical applications, and lessons learned for future stock assessment package development.

Another study by Punt et al. (2020b) recognized that integrated population modeling has become the preferred method for fish and invertebrate stock assessments, but current packages face limitations. Future stock assessment tools need to handle complex dynamics, scale from data-rich to data-poor scenarios, incorporate multi-species capabilities, and better manage temporal variations and tagging data. Essential features include efficient parameter sharing, comprehensive training programs, robust documentation, and flexible, modifiable design to adapt to evolving analytical needs.

Stock & Miller (2021) introduced the Woods Hole Assessment Model (WHAM) framework integrates empirical and mechanistic approaches to account for time-varying productivity in fisheries stock assessments. WHAM estimates random effects on numbers at age (NAA), natural mortality (M), and selectivity, and incorporates environmental covariates. Models with random effects, particularly those with a 2D AR (1) covariance structure, performed well in fitting data from diverse stocks, demonstrating negligible bias in key assessment outputs. WHAM shows promise for assessing stocks with variable productivity, such as U.S. Northeast groundfish, by incorporating time-varying processes and environmental links.

2.10 Case Studies and Applications of Length-based R Packages

Numerous case studies demonstrate the practical application of length-based R packages in fisheries management, highlighting their utility in diverse settings. Alam et al. (2022) assessed the stock status of Bombay duck (*Harpadon nehereus*) in Bangladesh, revealing overexploitation and low spawning biomass, and recommended specific mesh size and fishing pressure reductions. Similarly, Barua et al. (2022) used TropFishR and LB-SPR to evaluate the stock status of black pomfret (Parastromateus niger), which showed overexploitation and a low spawning potential ratio (SPR). Jiang et al. (2022a) used the LBB method to assess 26 fish species in a Chinese reserve, finding a significant "miniaturization" due to overfishing. These studies confirm that length-based methods are effective tools for evaluating stock health and guiding management decisions in data-limited fisheries. They also show the importance of species-specific evaluations that consider local ecological and management contexts.

Many case studies expose the degree of application of length-based R packages in fisheries management. Fish stocks in tropical areas have been evaluated, for example, using the "TropFishR" software, therefore generating information on recruitment trends, mortality, and growth rates (Mildenberger et al., 2017). These uses show the pragmatic value of length-based methods as well as the need of customized solutions for various fisheries. These case studies teach us the importance of ongoing improvement and validation of evaluation instruments to control newly developing problems (Pilling et al., 2008). The "fishmethods" package in temperate fisheries has also helped to better manage commercially valuable species by means of more exact stock estimations (Nelson, 2019).

Length-based R strategies have been extensively applied in many diverse fisheries all throughout the world. For instance, Sparre and Venema (1998) evaluated Caribbean demersal fish populations using length-based methods, therefore providing a valuable study for management. Taylor and Mildenberger (2017) first used "TropFishR" to assess coral reef fish counts in the Indo-Pacific region; its value in data-limited conditions was evident.

Furthermore, shown by case studies are the absolutely necessary tools for the sustainability of fisheries. By means of "TropFishR," Mildenberger et al. (2017) assessed Mediterranean economically important fish species, therefore providing guidelines for sustainable development strategies. Nelson (2019) also applied "fishmethods" techniques in the North Atlantic to improve groundfish stock estimate, so allowing more accurate management decisions.

Case studies also show how well length-based techniques mix with different kinds of data. To give a whole picture of tuna stocks in the Pacific, Maunder & Punt (2013) for instance coupled length frequency data with catch and effort statistics. Particularly in data-limited fisheries, this method has demonstrated to increase the accuracy and dependability of stock estimations (Pauly, 1987).

Jiang et al. (2022b) assessed 26 fish species in the Dongzhaigang National Reserve, Hainan Province, China, using CPUE and length-based Bayesian biomass (LBB) methods. Despite ambiguous CPUE results, LBB analysis showed significant "miniaturization" due to overfishing, with biomass ratios (B/B_0) below 0.5 for most species, indicating worsening overfishing since 2009. To sustain fisheries, local governments should collaborate with communities to reduce fishing effort, similar to past pollution reduction efforts.

Alam et al. (2021) reported the Hilsa shad (*Tenualosa ilisha*) fishery in Bangladesh, contributing 14% to the country's total fish production, is overfished due to increased fishing pressure. A rigorous assessment using three methods - traditional length-frequency based fishing mortality, Froese's length-based indicators (LBIs) for sustainability, and a surplus production-based Monte Carlo method (CMSY) - revealed overexploitation. Recommendations include implementing a minimum catch length of 33 cm, an 8 cm mesh size limit, and setting a yearly landing limit of 263,000–315,000 tons for sustainable management.

Medeiros-Leal et al. (2023a) addressed small-scale fisheries as so vital for food security, contributing to nearly half of global fish catches, but many of their stocks remain in poor condition. In data-limited scenarios, length-based methods are used to estimate stock status. This study applied three such methods length-based indicators (LBI), length-based spawning potential ratio (LBSPR), and the length-based Bayesian biomass approach (LBB) to assess the sustainability of fisheries in the Azores. The methods showed robustness for 15 out of 18 stocks, revealing that 45% of Azorean stocks are sustainable, 33% are in possible rebuilding/overfished status, and 22% are overfishing/overfished. Sensitivity analysis highlighted the impact of life-history parameter biases, with LBI being the most robust. Management recommendations include reducing catches and effort, increasing minimum landing and hook sizes for stocks needing rebuilding.

Barua et al. (2023a) studied the population dynamics and stock assessment of two commercially significant eel species in Bangladesh, *Muraenesox bagio* and *Congresox talabonoides*, using length-based and catch-based methods from July 2021 to May 2022. This study found that both eel stocks are overfished with spawning biomasses close to the limit reference points, recommending an optimal catch length and an annual landing limit of 250 mt to maintain stock biomass above BMSY levels.

Stock assessment using the length-based Bayesian biomass (LBB) technique was conducted Barman et al. (2021) for three sardine species in the Bay of Bengal, Bangladesh. The results showed that *Sardinella fimbriata* is overfished with a smaller-than-optimum length at first landing, while *Dussumieria acuta* and *D. elopsoides* have healthy biomass and safe fishing statuses. To ensure long-term viability, it is recommended to increase the mesh size of fishing gears.

The length-based Bayesian Biomass (LBB) approach was applied by Al-Mamun et al. (2021) to assess seven commercially valuable marine fish species (*Lepturacanthus savala*, *Pampus argenteus*, *Nemipterus japonicas, Nemipterus randalli*, *Ilisha filigera*, *Saurida tumbil*, and *Upeneus sulphureus*) in the northern Bay of Bengal, Bangladesh. This study revealed that five stocks are overfished, with only *Nemipterus japonicas* and *Nemipterus randalli* being healthy, and all species showed growth overfishing due to suboptimal capture sizes. This confirms the decline in Bangladesh's coastal water fishery resources. Specific targeted management measures are recommended to recover the country's marine fishery resources.

Abobi et al. (2019) investigated the stock status of cichlid species (*Oreochromis niloticus*, *Sarotherodon galilaeus*, and *Coptodon zillii*) in three reservoirs in northern Ghana using bootstrapping fish stock assessment (BFSA). Results indicate heavy exploitation but no alarming overexploitation, with *Sarotherodon galilaeus* having the highest biomass and *Coptodon zillii* the lowest. Length-based indicators show possible recruitment overfishing in some populations, necessitating prevention of increased fishing effort and further monitoring for improved management.

(Barua et al. (2023b) evaluated the stock status of *Penaeus monodon* (tiger shrimp), *Metapenaeus monoceros* (brown shrimp), and *Fenneropenaeus indicus* (white shrimp) in the Bay of Bengal by using length-based and catch-based methods. The relative biomass levels (B/BMSY) showed tiger shrimp were overfished (0.43), while brown (0.84) and white shrimp (0.96) were fully exploited but not overfished. The study recommends optimal carapace length limits for catching and estimates MSY reference points. Brown shrimp had the highest carrying capacity and intrinsic growth rate, indicating a better stock status compared to tiger and white shrimp.

Kindong et al. (2020) investigated the stocks of five non-targeted species in the Atlantic and Pacific Oceans by length-based Bayesian biomass estimate (LBB). The investigation found one population collapsed, three populations were overfished, and one was quite overfished. Two species in the Atlantic were greatly overfished; one species in the Pacific went extinct. Emphasized in the study is the need of improving fisheries data and monitoring the stock situation of non-target species.

Stock assessment for *Mullus barbatus* using catch-based (CMSY) and length-based (LBB, LBSPR, LIME) was carried out by Kılıç et al. (2023), data-limited models indicated a healthy stock status in 2020 with biomass above safe limits. The CMSY model showed catches did not exceed MSY, and LBB results indicated slightly higher fishing mortality than natural mortality. LBSPR and LIME results suggested sustainable biomass levels but highlighted the need for monitoring fishing pressure.

Raza et al. (2022) studied the stock status of five species from the marine water of Pakistan and reveled the marine fisheries resources of Pakistan have been significantly depleted, with four of five studied fish stocks showing overfishing and biomass levels below sustainable yield thresholds. The study recommends increasing mesh sizes and reducing the number of fishing boats to lower fishing mortality and ensure sustainability.

2.11 Future Directions and Innovations

The field of length-based stock assessment is continually evolving. Future directions include the development of more robust models that can account for multi-species interactions, environmental variability, and systematic observation errors (Stock & Miller, 2021; Dai et al., 2021a). The integration of hierarchical Bayesian models and machine learning algorithms into user-friendly packages is a crucial next step to provide more accurate and transparent assessments (Punt et al., 2020a). The emphasis will be on creating tools that are not only scientifically rigorous but also highly accessible, well-documented, and adaptable to a wide range of data-limited scenarios, which will ultimately support the sustainability of global fisheries.

Some of the developing trends in fish stock assessment are environmental data, advanced statistical techniques, and real-time monitoring systems (Hilborn & Walters, 1992). Innovations like automated data collecting via electronic monitoring (Punt et al., 2020b) and satellite data utilized for habitat mapping are helping assessments to get more accurate. New R packages should be developed with an eye toward adopting these advances, improving user-friendliness, and expanding the range of pertinent species and ecosystems (FAO, 2020). Cooperation among academics, managers, and developers will be very vital to forward these technologies (Hilborn et al., 2020). Moreover, artificial intelligence and machine learning models complicated interactions in marine ecosystems shows promise for more flexible and predictive management techniques (Dunn et al., 2019).

Recent study on the possibility of including environmental factors into length-based models to raise their predictive capacity Planque et al. (2010), for example, showed that adding temperature and other environmental variables into stock as assessment models might greatly increase their accuracy. Brander (2007) also underlined the requirement of using adaptive management techniques and the need of including climate change effects in stock evaluations.

Improved computational techniques have also created fresh opportunities for length-based stock assessment. Hierarchical Bayesian models, for instance, enable the integration of several sources of uncertainty, hence producing more reliable estimates (Gelman & Shalizi, 2013). Length frequency data has been subjected to machine learning techniques, which provide fresh understanding of stock dynamics and raise evaluation accuracy (Olden et al., 2008).

Another exciting direction of research is the creation of automated data collecting methods and real-time monitoring systems. For fishing boats, for example, electronic monitoring devices can offer precise and timely data on catch composition and size distribution, hence improving the quality of stock assessments (Karp et al., 2019). Using satellite data and remote sensing technologies, habitat maps and environmental condition monitoring is also under progress, therefore offering important information for stock assessments (Pikitch et al., 2004).

Hayashi et al. (2021) reported that as irregular production speeds could cause mistakes, the efficiency of stock assessment initiatives is absolutely vital. To simplify the workflow, a constantly integrated (CI) document system including cloud services for code hosting, computation, storage, and website hosting was used. This approach improved main tolerance, openness, and cooperation to make sure that changes and iterative work were properly included into reports. The Lean manufacturing method maximized the whole assessment process by increasing attention on necessary activities.

By matching applicable population dynamics models to fisheries, recording data, computing biomass and fishing mortality in respect to biological reference points, stock assessments help the decision-making process in fisheries. In a review study, Dichmont et al. (2016) looked at sixteen stock evaluation initiatives run in the United States. Even as they noted the challenges with usability and the evolution of auxiliary tools for more flexible packages, they highlighted the advantages of the software for researching assessment setups and facilitating peer review.

Table 2.1 Different R packages used for fish stock assessment

Package Name	Primary Developer(s)	Data for use (in addition to catch)	Primary references
Assessment Method for Alaska (AMAK)	Jim Ianelli	Age, Index	Anon (2015)
Age Structured Assessment Procedure (ASAP)	Chis Legault	Age, Index	Legault and Restrepo (1998)
Beaufort Assessment Model (BAM)	Erik Williams, Kyle Shertzer		Williams and Shertzer (2015)
Statistical Catch-At-Length (SCALE)	Paul Nitschke	Age, Size, Index	NOAA Fisheries Toolbox
Stock Synthesis	Richard Methot	Age, Length, Conditional age-at-length, Index, Discards, Tagging	Methot (1990); Methot and Wetzell (2013)
Simple Stock Synthesis (SSS)	Jason Cope	Age, Index	Cope (2013)
Virtual Population Analysis (VPA)	Many	Age, Index	Gavaris (1988); Conser and Powers (1990)
A Stock Production Model Incorporating Covariates (ASPIC)	Michael Prager	Index	Brodziak and Ishimura (2011); Brodziak et al. (2014)
Bayesian Surplus Production Model-2 (BSP2)	Mudoch McAllister, Beth Babcock	Index	McAllister (2014)
Depletion-Based Stock Reduction Analysis (DBSRA)	Ej Dick, Alec MacCall	Index	Dick and MacCall (2011)
FLR (Fisheries Library in R)	Iago Mosqueira and FLR Core Team	Catch, effort, bio-logical data	Kell et al. (2007)
SS3 (Stock Synthesis 3)	Richard Methot	Catch, effort, bio-logical, survey data	Methot & Wetzel (2013)

fishmethods	Marc H. Taylor	Various types (catch, biological, survey data)	Nelson (2015)
TropFishR	Daniel H. Ocampo, Rainer Froese	Length-frequency data	Mildenberger et al. (2017)
LB-SPR (Length-Based Spawning Potential Ratio)	Timothy R. Carruthers, Carrie R. Brown	Length-frequency data	Froese et al. (2018)
LBB (Length-Based Bayesian Biomass)	Rainer Froese, Armand M. E. Gascuel	Length-frequency data	Froese et al. (2018)
JABBA (Just Another Bayesian Biomass Assessment)	Henning Winker, Andre E. Punt, Alistair J. Hobday	Catch and CPUE data	Winker et al. (2018)

CHAPTER 03 METHODOLOGY

3 METHODOLOGY

3.1 Developing Approach and Description of the aLBI Tool

3.1.1 Package development

Package development in R empowers researchers and analysts to create reusable, efficient, and shareable code tailored to specific analytical needs. By packaging functions, data, and documentation, users can streamline workflows and ensure reproducibility. This approach enhances collaboration and allows for easy distribution of tools within the R community.

3.1.2 Creating the aLBI package

A simple non-overlapping unique name *aLBI* was given for the package which defines it best. This makes it easy for potential users to find your package and associated resources. also check if a name is already used on CRAN by loading http://cran.r project.org/web/packages/[aLBI]. The aLBI package was created by following steps shown in Figure 3.1. The package was created with a D/R/MyRpackage directory, man/directory, basic *DESCRIPTION*, *NAMESPACE* file etc. shown in Figure 3.1. The R project will also include an RStudio project file aLBI.Rproj, which makes aLBI package easy to use with RStudio. A GitHub repository was also created for the hosting of this package and to make available all the updates.

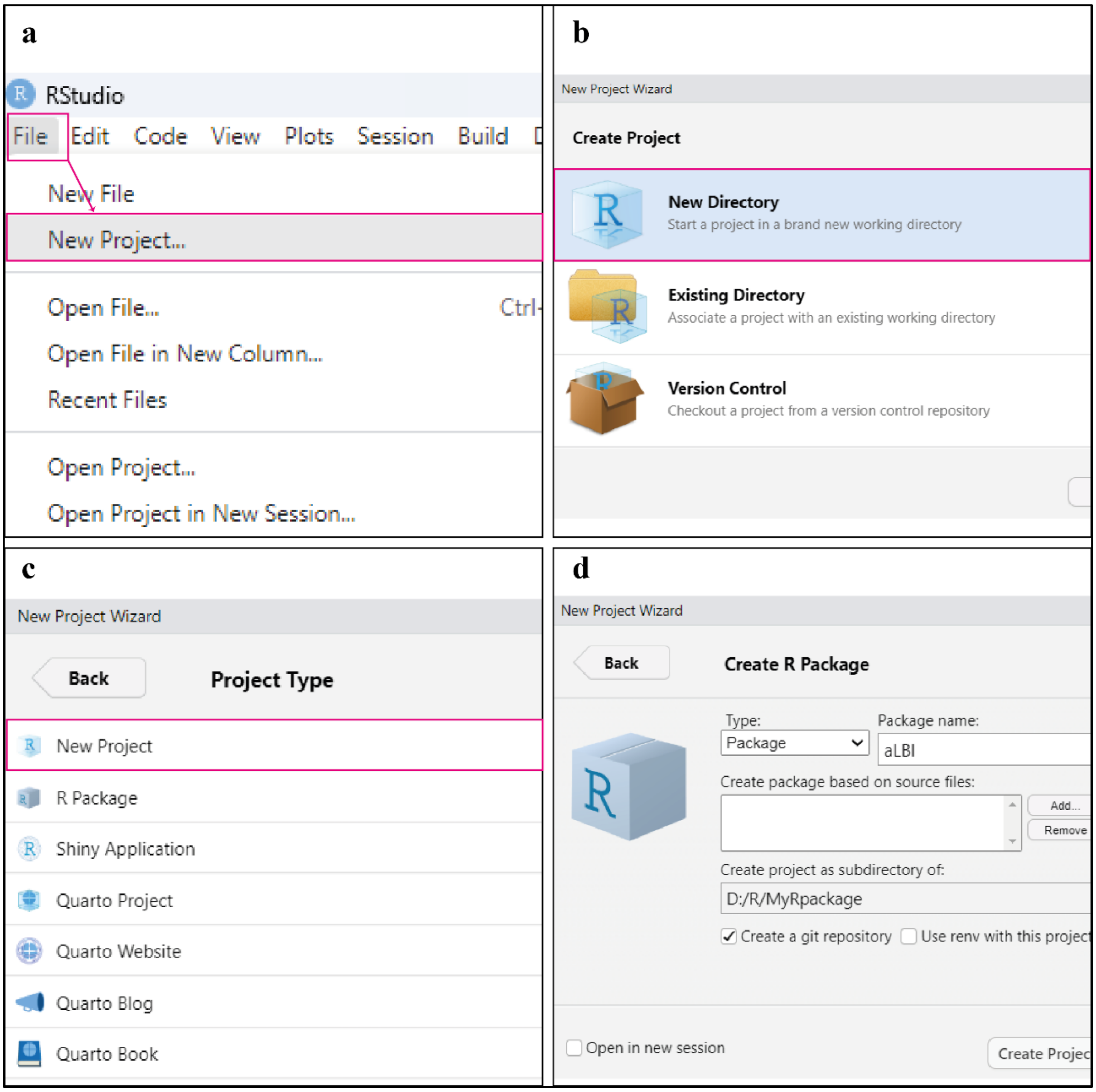

Figure 3.1: Steps followed in the creation of aLBI package. Creating a new project from the file menu (a), creating a project from a new directory (b), creating a new R package (c), naming the R package and creating the project with git repository (d).

3.1.3 Study Area Selection

This study focuses on the collection of length frequency data of *S. aor* from Kaptai Lake, located in the Chattogram district of Bangladesh. The lake lies in the southeast part of Bangladesh, within the coordinates of latitude 22°22' to 23°18'N and longitude 92°00' to 92°26'E (Figure 3.2). Kaptai Lake, the largest man-made freshwater lake in the country with a water area of 68,800 hectares which is 1.46% of the total inland open water bodies (DoF, 2023), is a vital resource for local fisheries and communities. The lake interconnects several upazilas, including Rangamati Sadar, Langadu, Nannerchar, Jurachari, Kaptai, and Belaichari. It is fed by the Karnaphuli, Chengi, Mayani, Kassalong, and Rinkhyong rivers. Fishing activities in Kaptai Lake play a crucial role in the livelihoods of local communities and contribute significantly to Bangladesh's inland fisheries production (Galib et al., 2018; Suman et al., 2021; Shalehin et al., 2022). As the largest artificial lake in the country, Kaptai Lake provides an expansive fishing ground, supporting both commercial and subsistence fishing practices (Rahman et al., 2024). The lake is home to a wide variety of freshwater fish species, including native species like catfish (*Sperata aor*, *Mystus spp.*) and carps (*Labeo rohita*, *Cirrhinus mrigala*), as well as introduced species like tilapia (*Oreochromis spp.*) (Lima et al., 2023).

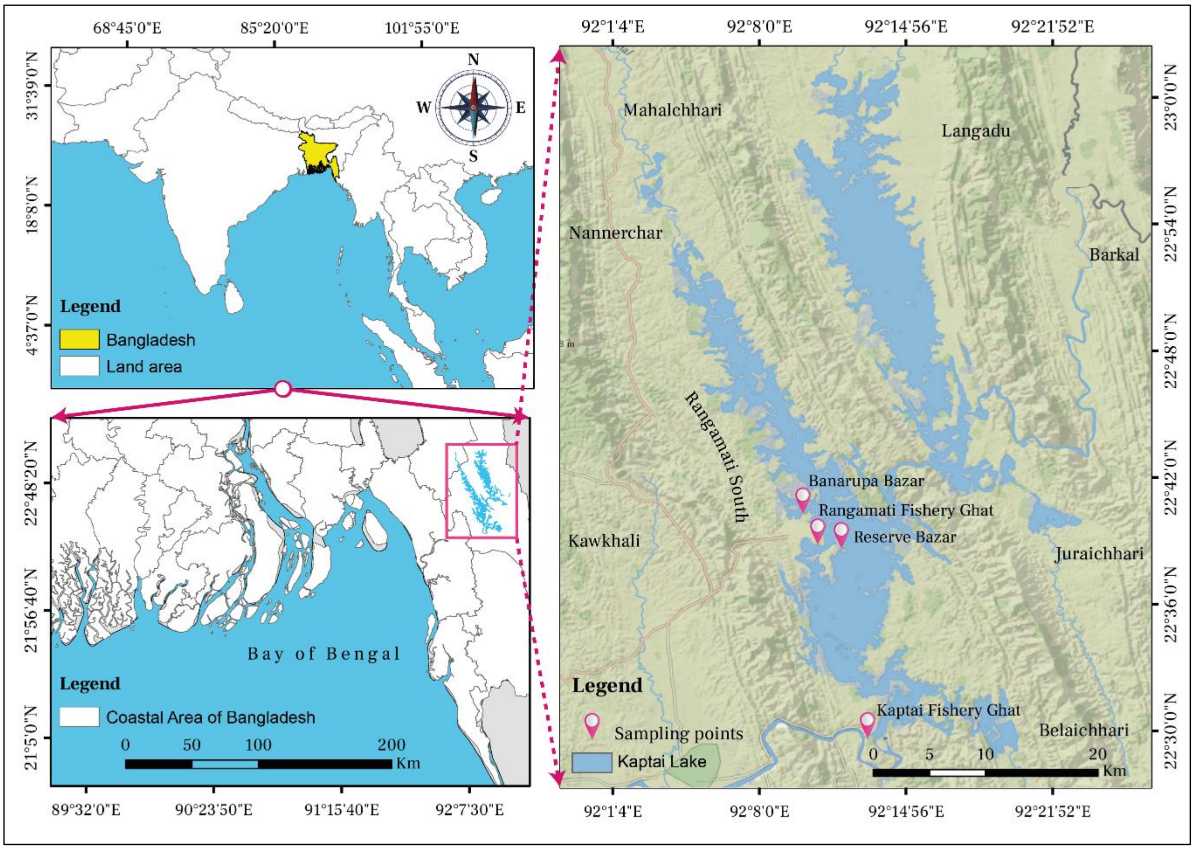

Figure 3.2: Map showing the sampling points of the study

Kaptai lake fisheries multi-species and multi-gear in nature (Rahman et al., 2024) where fishing methods employed in range from traditional techniques such as cast nets, gill nets, traps, and seine nets to modern fishing gear for commercial operations. Seasonal fishing activities, heavily influenced by monsoon patterns, are vital for meeting the local demand for protein and generating income. However, overfishing, habitat degradation, and water level fluctuations pose challenges to the sustainability of these activities. Efforts to manage Kaptai Lake's fisheries include community-based management and government interventions to ensure stock sustainability. Notably, seasonal fishing bans (April to July) and the establishment of conservation zones are key strategies implemented to maintain fish biodiversity and regulate fishing activities. These approaches aim to balance ecological preservation with the socio-economic needs of local communities. (Pearce et al., 2002; Suman et al., 2021; Shalehin et al., 2022)

3.1.4 Target Species: *Sperata aor*

The *Sperata aor* (Hamilton, 1822), commonly known as the long-whiskered catfish, is an important species within the Bagridae family, widely distributed across freshwater systems in South Asia, including the river basins of the Ganges, Brahmaputra, and Indus. In Bangladesh, this species is notably prevalent in Kaptai Lake, an artificial reservoir created in 1966 by damming the Karnaphuli River, is an important habitat for almost all indigenous fish species in Bangladesh. Recognized as a hotspot for biodiversity, this lake plays a critical role in providing food, nutrition, and livelihoods for thousands of local people (Hossain et al., 2012). *S. aor* significantly contributes to the annual total landings and is one of the major fish species harvested from the lake. (Shalehin et al., 2022). However, the species is experiencing increasing fishing pressure driven by the rising demand for indigenous fish at both local and national levels. Despite its importance, no comprehensive study has been conducted to assess the stock status of *S. aor* in Kaptai Lake. Studies on other species in the lake have revealed signs of overexploitation (Bashar et al., 2021; Khatun et al., 2023; Mia et al., 2024), raising concerns about similar trends for *S. aor*. Given its ecological and economic significance (Rana et al., 2024), evaluating the current stock status of *S. aor* is imperative. Such an assessment is essential for developing effective conservation measures to ensure sustainable exploitation, maintain ecological balance, and preserve the biodiversity that supports the local economy.

3.1.5 Collection of length frequency data and processing

Four sampling stations (Figure 3.2): BFDC Fishery Ghat, Rangamati (22°38'55.22"N & 92°11'4.41"E); BFDC Fishery Ghat, Kaptai (22°29'51.12"N & 92°13'11.66"E); Banarupa Bazar (22°39'13.36"N & 92°10'44.15"E); and Reserve Bazar (22°39'16.26"N & 92°11'49.33"E) were selected for monthly length data collection from August 2022 to March 2023 (Figure 3.3). Due to the fishing ban between April and July, data were not collected in this period. In the landing stations, fisher gathers their catches in different pile. During the high landings, fishers were asked to mixed the pile properly and 10% of the well-mixed pile were collected for collecting the length data. Similarly, during the low landings, 30 – 100% fish were collected. Lengths of the fish were measured using a meter scale to the nearest 0.1 cm using a measuring scale and recorded in excel file in s single column to compiled into a two-column data frame (Length and Frequency) of different bin size using *FrequencyTable* function of aLBI package. Data quality was validated using *FishPar* function within the aLBI package, ensuring positive frequencies, a minimum sample size of 100 individuals, and at least three unique length classes. Warnings were issued for skewed distributions (e.g., >80% frequency in a single length class) to flag potential biases that could affect parameter estimation.

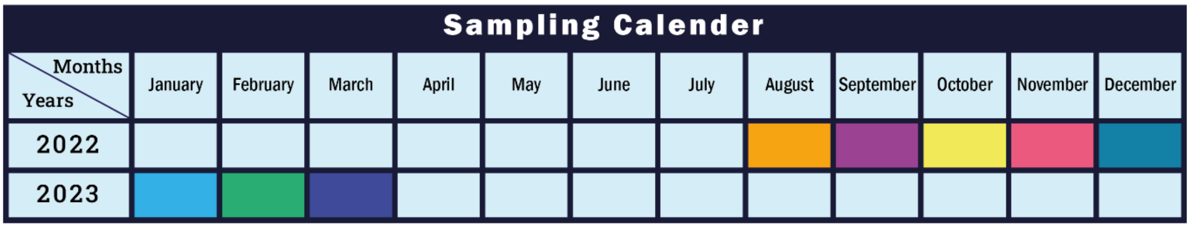

Figure 3.3: Sampling calendar of the study

3.1.6 Package Development and Documentation

The development of the aLBI package is meticulously executed, adhering to established coding standards and best practices to ensure robustness and efficiency. Each function and module within the aLBI package is carefully crafted, tested, and optimized for performance. Comprehensive documentation is produced to accompany the development process (Figure 3.4), providing users with clear and detailed instructions on installation, configuration, and usages.

This documentation includes thorough descriptions of all functions, classes, and modules, accompanied by practical examples and use cases were automatically created with the *roxygen2* (Wickham et al., 2024a) packge. The package's comprehensive documentation includes a detailed README file, which offers an overview of the package, installation instructions, and a quick start guide. Vignettes are also provided, offering in-depth tutorials and real-world examples that showcase the package's capabilities. These vignettes are particularly useful for new users, as they provide step-by-step instructions on how to leverage the package for various analyses. Additionally, it details the dependencies and requirements needed for the package, ensuring users have all necessary information to integrate and utilize the package effectively. Regular updates and version control are maintained to keep the documentation current with any changes or improvements, ensuring users always have access to the latest features and fixes

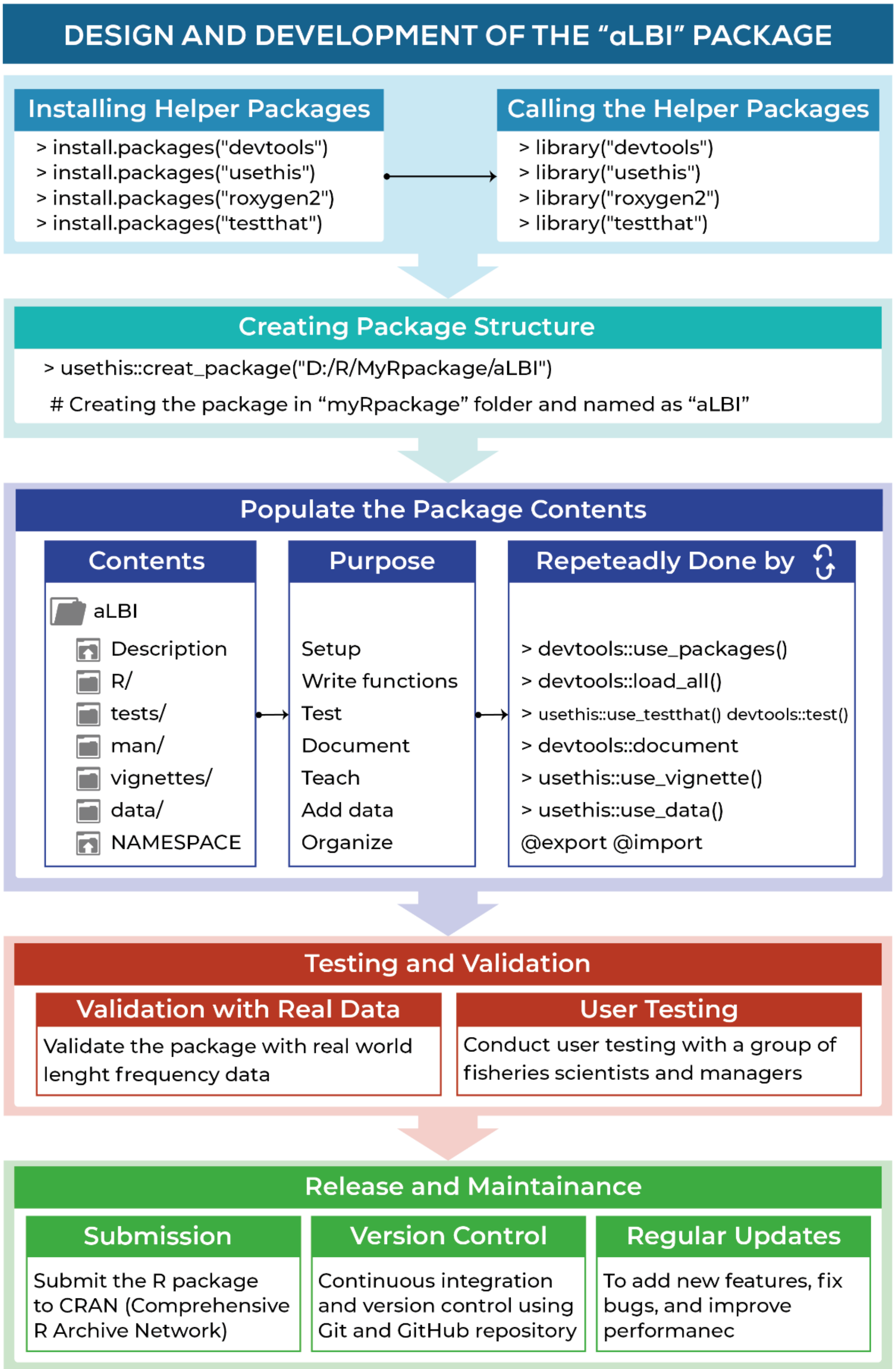

Figure 3.4: Design and development process of the aLBI package

3.1.7 Development and planning of functions

The development and planning process of aLBI package involves a repeated process from uncountable times (Figure 3.5). The first practical advantage to using a package is that it's easy to reload all the code and functions. All the open files were saved by running devtools::load_all() and also whenever it needs to change or edit any code or functions, this workflow will reload all the edited R files (Figure 3.5).

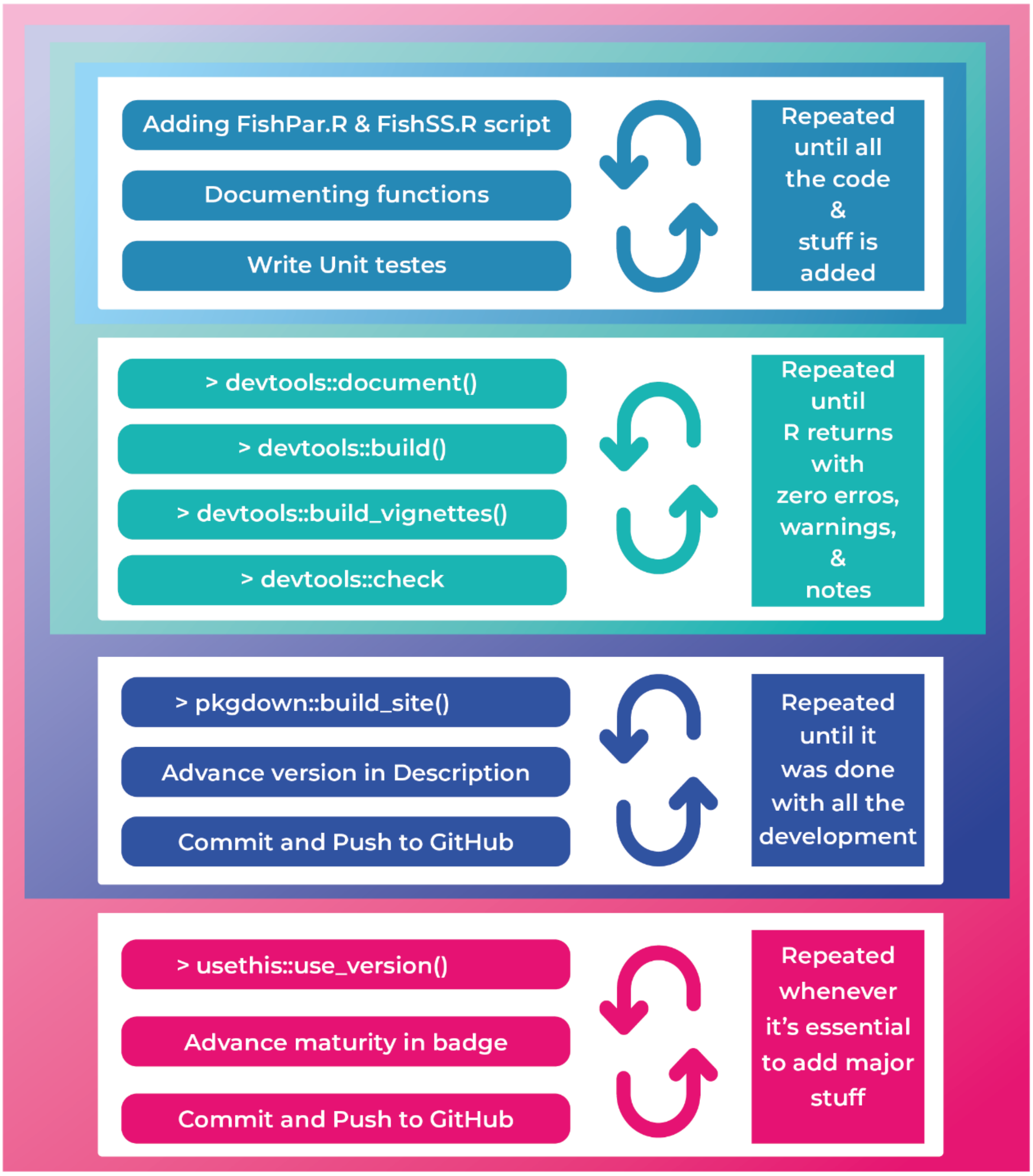

Figure 3.5: Development and planning process of the functions of aLBI package

There are two R files containing two main functions named as FishPar and FishSS were (Figure 3.5) designed to estimate some of the key biological parameters (Lmax, L_∞, Lmat, Lopt, LM_ratio, Froese length indicators (Pmat, Popt, Pmega), stock. The *FishPar* function processes length frequency data, performs bootstrap resampling to calculates key biological parameters, and generates corresponding plots (histograms and density plots, boxplot) for graphical visualization of the estimated parameters. On the contrary, *FishSS* function evaluates stock status (*TSB40* and *LSB25*) based on length-based indicators (Pobj, Pmat, Popt, LM_ration) by selecting specific columns and then values (Figure 3.6).

The *FishPar* function facilitates this by estimating key biological parameters, such as the maximum length (L_{max}), asymptotic length (L_{∞}), and length at maturity (L_{mat}), using Montecarlo simulation. It ensures data integrity by validating input data and generating robust confidence intervals for parameter estimates. Additionally, it provides insightful visualizations to help interpret the results, making it a valuable tool for researchers and fishery managers.

In addition, the *FishSS* function analyzes particular indicators such as *LM_ratio* and P_{obj} to ascertain the stock situation. Analyzing these indications helps one to choose suitable columns from the input data to create important stock metrics such Low Spawning Biomass (*LSB25*) and Total Spawning Bio mass (*TSB40*). This dual-function strategy improves our knowledge of fish population dynamics, therefore helping to guide the development of sensible management plans and fisheries policies. These purposes together promote the aim of sustainable fisheries management by incorporating statistical rigor with practical application (Figure 3.6).

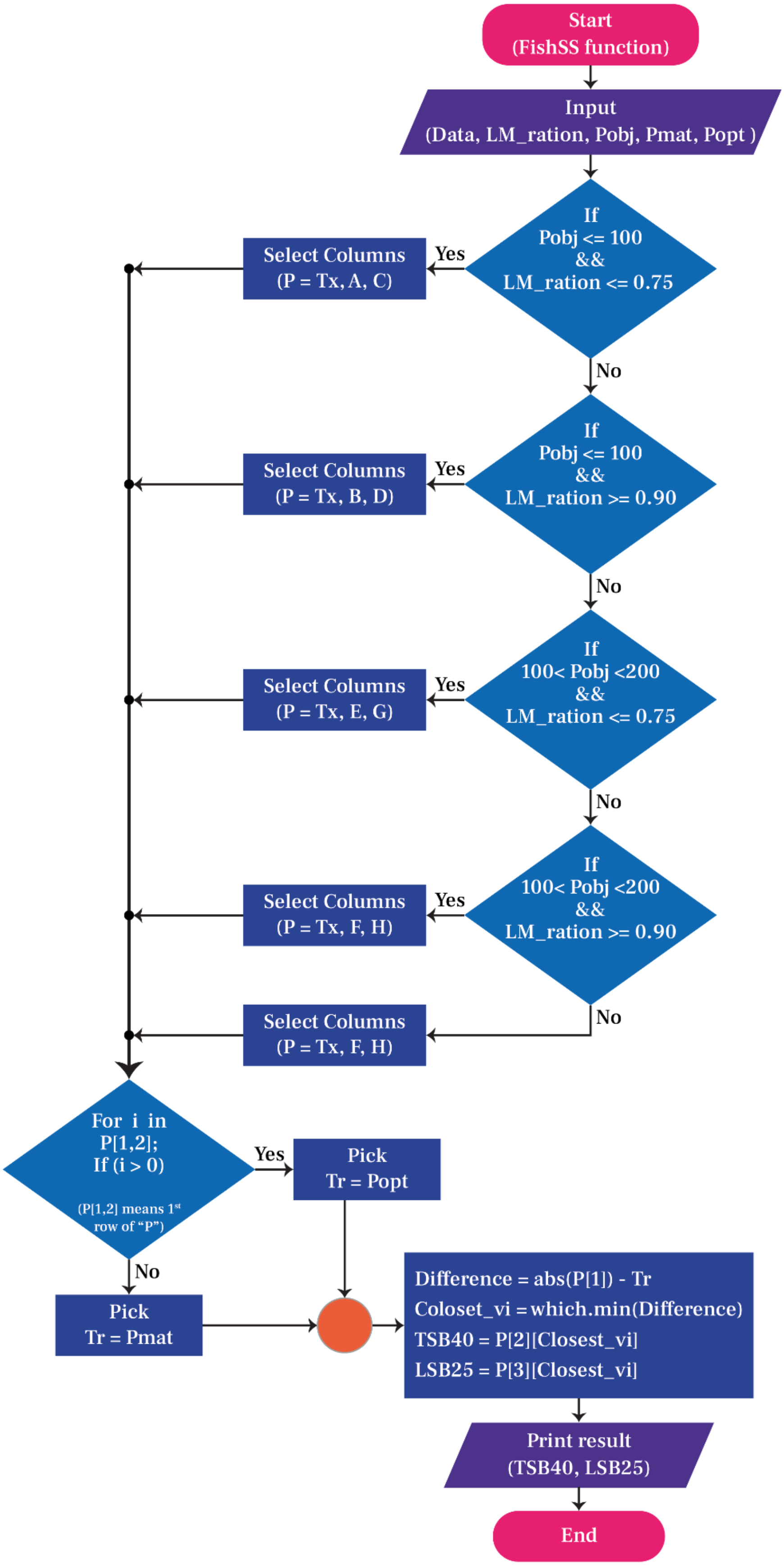

Figure 3.6: Flow chart of the FishSS function which finds out the stock status.

3.1.8 Package Details and Software Dependencies

Version 0.1.9 of the *aLBI* package is written in the R programming language (version 4.5.1) and is organized using the *roxygen2* documentation system (Wickham et al., 2024a). This package is designed to function seamlessly within the R environment (R>= 4.0.0) and depends on a set of existing packages to perform its operations. These dependencies, listed in the *DESCRIPTION* file, include *dplyr* (Wickham et al., 2024b) for data manipulation, *openxlsx* (Schauberger et al., 2025) for reading and writing Excel files, *stats* (WickBolar, 2019) and *Rgraphics* (Murrell, 2020) for core statistical functions and plotting, *grDevices* (Bengtsson, 2024) for graphical device control, *ggplot2* (Wickham et al., 2025a) for data visualization, *rlang* (Henry et al., 2024) for extracting upper length class to construct frequency table, and *utils* for various utility functions.

The aLBI R package was developed using a test-driven development (TDD) approach, which prioritizes the creation of unit tests before writing the main code. Unit tests were implemented using the *testthat* package (Wickham et al., 2025b), a key dependency for testing. These tests were conducted on mock length-frequency data, which was populated with synthetic test-specific data to ensure the package accurately encoded all software requirements. For example, tests were developed to check the output format of results from core functions like *FrequencyTable*(), *FishPar*(), and *FishSS*(), as well as to perform numerous logical checks on the results to ensure their accuracy (Figure 3.4). Furthermore, rigorous tests were implemented to ensure that edge cases, such as handling data with specific biases or extreme values, were correctly managed. This process also confirmed that expected errors would return informative and helpful error messages to the user, enhancing the package's user-friendliness. Once all unit tests were passing, the code was then systematically refactored to optimize for computational speed and memory usage, ensuring a robust and efficient final product.

3.1.9 Installation of the Package

The package (aLBI) is accessible on GitHub repository (for latest update) and also available on CRAN (https://cran.r-project.org/web/packages/aLBI/index.html). For detailed installation instructions, troubleshooting tips, and additional information, users can refer to the vignettes (https://cran.r-project.org/web/packages/aLBI/vignettes/Introduction.html).

Users should first make sure R installed on their system corresponds with their preferred version 4.0.0 or above. Users of the devtools (Wickham et al., 2022) package in R for installations from GitHub. Check first that *devtools* is loaded and installed by running:

> install.packages("devtools")

> library(devtools)

Next, install the aLBI package directly from the GitHub repository using the following command:

> install_github("Ataher76/aLBI")

This command will download and install the aLBI package from the specified GitHub repository. If any dependencies are required, they will be installed automatically during this process.

For users who prefer to wait for the CRAN release, they can install the package directly from CRAN using the standard installation command:

> install.packages("aLBI", dependencies= TRUE)

3.1.10 Main functions of aLBI package

Table 3.1 Main Functions of aLBI R package

Functions	Descriptions	Usage	Arguments with Description
Frequency Table	This function creates a frequency distribution table for fish length data	FrequencyTable(data, method, bin_width, Lmax, output_file)	***data*:** A numeric vector of fish lengths. ***bin_width:*** Optional. A numeric value specifying the bin width for class intervals, if left NULL, the bin width is automatically calculated using Wang et al. (2020) optimum bin size (OBS) formula. **L_{max}:** Optional. The maximum length of fish. If left NULL then automatically calculate the frequency table with the observed maximum length of the provided data. **output_file:** Character string specifying the output Excel file name. Defaults to "FrequencyTable_Output.xlsx".
FreqTM	Creates frequency distribution table for multiple months	*FreqTM* (data, bin_width, Lmax, date_config, output_file)	***Data***: A data frame containing columns for months and lengths. **bin_width**: Numeric value specifying the bin width for class intervals. If NULL (default), bin width is calculated using Wang's formula. **Lmax**: Numeric value for the maximum observed fish length. Required only if `bin_width` is NULL and Wang's formula is used. Defaults to NULL.

Functions	Descriptions	Usage	Arguments with Description
			date_config: A list with elements `day` (default 1) and `year` (default 2025) to set the day and year for converting month names to dates. The day must be between 1 and 31. **output_file**: Character string specifying the output Excel file name. Defaults to "FreqTM_Output.xlsx".
FishPar	Calculates length-based indicators using Monte Carlo simulation for length parameters and non-parametric bootstrap for Froese indicators.	FishPar(data, resample, progress, Linf, Linf_sd, Lmat, Lmat_sd)	***data***: A data frame containing two columns: Length and Frequency. ***resample***: An integer indicating the number of bootstraps resamples. ***progress***: A logical value indicating whether to display progress. ***L_∞:*** A numeric value for the asymptotic length (optional). ***Linf_sd***: A numeric value for the standard deviation of random variation added to Linf (default: 0.5). Only used if L_∞ is provided by users. ***L_{mat}:*** A numeric value for the length at maturity (optional). ***Lmat_sd*** A numeric value for the standard deviation of random variation added to Lmat (default: 0.5). Only used if *Lmat* is provided by users.
FishSS	Assesses the stock status based on parameters calculated by the FishPar function	FishSS(data, *LM*_ratio, P_{mat}, P_{opt}, P_{mega})	**data:** A data frame containing the necessary columns for stock status calculation. (Cope & Punt, 2009) ***LM*_ratio**: A numeric value representing the ratio of L_{mat} and L_{opt}. **P_{mat}**: A numeric value representing the percentage of mature fish. **P_{opt}**: A numeric value representing the percentage of optimally sized fish. **P_{mega}**: **A** numeric value representing the percentage mega-spawner.
LWR	Visualizes and models the relationship between length and weight (or any two continuous	LWR(data, log_transform, point_col, line_col, shade_col, point_size, line_size, alpha,	**data** A data frame with at least two columns: the first for length, the second for weight. **log_transform**: Logical. Whether to apply a log-log transformation to the variables. Default is TRUE.

Functions	Descriptions	Usage	Arguments with Description
	variables) using linear regression.	main, xlab, ylab, save_output)	**point_col:** Color of the data points. Default is black. **line_col:** Color of the regression line. Default is red. **shade_col:** Color for the confidence interval ribbon. Default is red. **point_size:** Size of the data points. Default is 2. **line_size:** Size of the regression line. Default is 1. **alpha**: Transparency for the confidence interval ribbon. Default is 0.2. **main:** Title of the plot. Default is Length-Weight Relationship. **xlab** Optional. Custom x-axis label. **ylab:** Optional. Custom y-axis label. **save_output:** Logical. Whether to save the plot as a PDF and the model summary as a text file. Default is TRUE.

The *FrequencyTable* function generates the frequency table as well as provides a data frame containing the upper length class and their corresponding frequencies. With this frequency table the *FishPar* function facilitates the analysis by estimating key biological parameters, such as the maximum length (L_{max}), asymptotic length (L_∞), and length at maturity (L_{mat}), using Monte Carlo and bootstrap methods (Table 3.1). It ensures data integrity by validating input data and generating robust confidence intervals for parameter estimates. Additionally, it provides insightful visualizations to help interpret the results, making it a valuable tool for researchers and fishery managers. In addition, the *FishPar* function also calculate particular indicators such as L_{mat}/L_{opt} ratio (*LM*_ratio) and sum of P_{mat} P_{opt} P_{mega} (P_{obj}) to ascertain the stock status. Analyzing these indicators help one to choose suitable columns from the input data to create important stock metrics such as Limit Spawning Biomass (*LSB25*) and Target Spawning Biomass (*TSB40*) (Table 3.1) by *FishSS* function. This dual-function strategy improves our knowledge of fish population dynamics, therefore helping to guide the development of sensible management plans and fisheries policies.

3.1.11 Testing and Validation with Real World Data

The testing and validation technique of the aLBI package guarantees its accuracy, dependability, and efficiency in pragmatic usage by a rigorous approach. First verified unit-wise, the package guarantees that every component is functionally sound. These tests seek to evaluate, under several scenarios and with various inputs, the precision of every function and module (Figure 3.4). After that, the package is integrated and tested closely to make sure every component runs correctly taken together.

Using the *aLBI* program on simulated length frequency data helps to authenticate real-world usage scenarios by means of actual data replication. Moreover, under close inspection is the initiative of managers leveraging own data and fisheries experts. The performance of the program is evaluated using particular criteria combining pre-calculated figures from an Excel file with accuracy, precision, recall, and computation efficiency. Any flaws discovered at this phase are extensively investigated and fixed repeatedly.

To assess its performance and stability under demanding loads and hostile environment, the package also passes extensive stress tests. Furthermore, learned from real-world implementations is gathered to advance the project. The rigorous testing and validation approach guarantees the aLBI package to be sturdy, trustworthy, and ready for operation in reasonable, real-world settings (Figure 3.4).

3.1.12 Publishing and Maintenance

Maintaining and releasing the aLBI package requires for a comprehensive methodology to guarantee end-user availability, usability, and dependability. First developed and tested extensively to meet CRAN's standards, the package is the package is submitted in to CRAN for review when completed. This submission consists of the package source, documentation, and samples proving their use. Every comment made by CRAN maintainers is given immediate attention during the review process to adhere to their policies.

The package is also accessible on GitHub concurrently via the link https:// github.com/Ataher76/aLBI. By means of continuous integration and test runs, this platform allows users to report issues, provide ideas for enhancements, and assist the package to be developed. Git guarantees that the most recent version is always available by regular updates and bug fixes sent to the GitHub repository (Figure 3.4).

Maintenance of the package is an ongoing effort. It involves monitoring for any reported bugs, addressing compatibility issues with new versions of R or other dependencies, and incorporating user feedback to improve functionality. Git commit and push versioning system is applied to manage releases, with major, minor, and patch versions clearly defined. This systematic approach ensures that the aLBI package remains robust, user-friendly, and reliable for all its users. Additionally, documentation is kept up-to-date to reflect any changes or additions to the package features.

3.2 Analytical procedures

3.2.1 Calculating frequency table by FrequencyTable function

Frequency distribution table was constructed using the following formula of optimum bin size (OBS) recommended by Wang et al. (2020), facilitate by the *Frequency Table* function of aLBI package.

Equation 1: Formula of Calculating Optimum Bin Size (OBS)

$$\text{Optimum Bin Size (OBS)} = (0.23 * (Lmax^{0.6}) \text{..............} (1)$$

L_{max} = Maximum length

3.2.2 Estimating growth parameters by FishPar function

Key length parameters for *S. aor*, including maximum length (L_{max}), asymptotic length (L_{∞}), length at maturity (L_{mat}), and optimum length (L_{opt}), were estimated using the *FishPar* function within the aLBI package. L_{max} was defined as the maximum observed length in the dataset unless derived from a user-specified L_{∞}. When not provided, L_{∞} was calculated according to following equation proposed by Pauly (1984).

Equation 2: Formula of calculating asymptotic length (L_{∞})

$$L_{\infty} = \frac{L_{max}}{0.95} \text{..} (2)$$

The L_{mat} and L_{opt} for *S. aor* were estimated using empirical equations from Froese & Binohlan (2000) within the FishPar function of the aLBI package. The package used the following equation to calcualte L_{mat}.

Equation 3: Formula of Calculating Length at Maturity (Lm)

$$\log(L_{\text{mat}}) = 0.8979(\log(L_{\infty})) - 0.0782 \text{...............} (3)$$

Where, L_{∞} is the asymptotic length.

After the estimation of L_{mat}, the automated procedure of the package estimated the L_{opt} using the following formula:

Equation 4: Formula of Calculating Optimum Length (L_{opt})

$$\log(L_{\text{opt}}) = 1.053(\log(L_{\text{mat}})) - 0.0565 \text{...............} (4)$$

The *FishPar* function allowed user-specified values for L_{∞}, and L_{mat} with default standard deviations (*Linf_sd = 0.5* and *Lmat_sd = 0.5*) or user-defined variability to reflect uncertainties from sources like FishBase or empirical studies. When parameters were estimated from data, a default standard deviation of 0.015 was applied to log-transformed L_{mat} and L_{opt} to account for variability. The inclusion of these parameters was a direct response to the need to explore sensitivity in parameter estimates, ensuring the model could simulate realistic variability in growth and maturity metrics.

3.2.3 Monte Carlo Simulation for Parameter Uncertainty

To quantify uncertainty in length parameters (L_∞, L_{max} L_{mat}, L_{opt}), the package performed Monte Carlo simulations using the FishPar function, generating 1,000 simulated datasets. L_{max}*_samples* were drawn from a normal distribution centered on the observed L_{max} with a standard deviation of 1 cm, constrained to 90–110% of L_{max} for biological plausibility. L_∞ samples were derived as L_{max}*_samples* / 0.95 (Pauly, 1984) or from user-specified L_∞ with added noise (normal distribution, mean = 0, sd = *Linf_sd* = 0.5 cm). L_{mat} and L_{opt} samples were calculated using empirical equations from Froese & Binohlan (2000) with added noise (standard deviation = 0.015 for log-transformed values). Results were stored in a *parameter_estimates* matrix, with means and 95% confidence intervals computed using the 2.5th and 97.5th percentiles via R's stats package (version 4.5.1). This approach ensured robust parameter estimates by accounting for inherent biological variability, as recommended by Thorson et al. (2014a).

3.2.4 Estimating Froese Length-Based Indicators (LBIs) by FishPar function

In order to evaluate the stock status in relation to exploitation and prevent excessive fishing that could harm development and recruitment, Froese (2004) suggested three length-based indicators (LBIs) known as P_{mat}, P_{opt}, and P_{mega}.

1. The first indicator P_{mat}, can be defined as the proportion of mature fish in the catch, and it represents the concept of "let them spawn!" The percentage of fish in the capture that have a length larger than the length at sexual maturity (L_{mat}) was determined using the following formula proposed by Froese (2004):

Equation 5: Formula of calculating percentage of mature fish (Pmat)

$$P_{mat} = \sum_{L_{mat}}^{L_{max}} \mathrm{P}_L \text{.....................} (5)$$

Where, P_L is the percentage of fish in the catch composition with a length interval of L.

2. Indicator two P_{opt}, defined as the percentage of fish caught at their optimal length (L_{opt}), which is the length that maximizes the yield and revenue.. P_{opt} was calculated by following formula:

Equation 6: Formula of calculating percentage of optimally length fish (Popt)

$$P_{opt} = \sum_{0.9L_{opt}}^{1.1L_{opt}} \mathrm{P}_L \ldots\ldots\ldots\ldots\ldots (6)$$

3. The P_{mega}, defined as the percentage of old and larger fish (mega = mega spawner) in the catch composition. The metric is calculated as the proportion of mature fish in the catch that exceed the optimal length by 10%. The following equation was used to estimate P_{mega}:

Equation 7: Formula of calculating percentage of mega-spawners (Pmega)

$$P_{mega} = \sum_{1.1L_{opt}}^{L_{max}} \mathrm{P}_L \ldots\ldots\ldots\ldots\ldots (7)$$

3.2.5 Non-Parametric Bootstrap for LBI Uncertainty

To quantify the uncertainty in LBIs, the package used a non-parametric bootstrap approach within the FishPar function. Length-frequency data were expanded into individual observations based on frequencies, and 1,000 bootstrap samples were generated by resampling with replacement, preserving the total frequency. For each sample, P_{mat}, P_{opt}, and P_{mega} were calculated using length ranges derived from Monte Carlo simulations. The 2.5th and 97.5th percentiles of the resulting distributions were used to compute 95% CIs, with bounds capped at 0% and 100% to ensure biological plausibility. This method, implemented using R's sample and quantile functions, provided robust estimates of indicator variability, complementing the Monte Carlo approach for length parameters (Efron, 1992).

3.2.6 Stock spawning biomass (SB) evaluation

FishSS function evaluates the stock status using a decision table (*CPdata*) demonstrated by Cope and Punt (2009), which is readily available within the aLBI package based on the input parameters (*LM_ratio*, P_{mat}, P_{opt}, P_{mega}) derived from FishPar function. The function classified stock health relative to limit (*LSB25*, 25% of unfished spawning biomass) and target (*TSB40*, 40% of unfished spawning biomass) reference points to determine whether the stock was overfished or sustainable. P_{mat} aimed to ensure all fish reproduce at least once before capture (target: 100%), P_{opt} targeted capture within ±10% of the optimum length ($0.9L_{opt}$ to $1.1L_{opt}$, target: 100%), and P_{mega} indicated a healthy stock structure at 30–40% or potential overexploitation below 20% (Froese, 2004). These metrics, integrated into the decision table, provided robust insights for fisheries management.

3.2.7 Evaluating the robustness of the aLBI package

To evaluate the robustness of the aLBI package under varying bin sizes, the length frequency data were rearranged using the *FrequencyTable* function of the aLBI package with bin sizes of 1 to 5 cm, in addition to the base model with a bin size of 3 cm. Examining bin structure is crucial because it influences the resolution of length-frequency distributions, thereby affecting the estimation of length parameters (e.g., L_{max}, L_{∞}, L_{mat}, and L_{opt}) and the accuracy of derived length-based indicators (LBIs), while potentially masking biological patterns or introducing bias if bins are too coarse or too fine. The uncertainty in length-based indicators (LBIs) (P_{mat}, P_{opt}, and P_{mega}) was propagated from the uncertainty in key length parameters such as length at maturity (L_{mat}), optimal length (L_{opt}), and maximum length (L_{max}). The uncertainty in these length parameters was estimated through a bootstrapping approach, which resampled the length data 1,000 times to generate confidence intervals for each parameter. The LBIs for each bin size were then estimated using the outputs of the bootstrapping procedure. Thus, the inherent uncertainties in the length data were addressed to produce robust estimates of the fishery. To test the significant difference among the length parameters and LBI estimates, one-way ANOVA and Turkey's HSD test were also employed.

3.2.8 Sensitivity Analysis of aLBI R Package

To evaluate the robustness of the aLBI R package, a comprehensive sensitivity analysis was performed on the key user defined input parameters (L_{∞}, L_{mat}). This analysis focused on the length parameters (L_{∞}, L_{mat}, L_{max}, L_{opt}, L_{opt_m10}, and L_{opt_p10}) and length-based indicators (P_{mat}, P_{opt}, P_{mega}) estimated by *FishPar* function. This approach involved systematically varying each of these two parameters by ±5% while holding all other parameters at their base model values. This analysis was designed to reflect uncertainties and real-world variability beyond typical bounds, thereby providing a more comprehensive understanding of the package's sensitivity. A total of nine scenarios were generated (a base model plus eight scenarios from the ±5% variation of the four input parameters), as summarized in Table 4.4. This allowed for a robust overview of the model's performance under data-limited conditions, demonstrating how changes in a single parameter influence the model estimations.

3.2.9 Assumptions and Validation

The methodology assumed stock equilibrium (stable growth, mortality, and recruitment), fisheries-representative sampling, and consistent fishing pressure, as required for LBIs (Froese, 2004). These assumptions can be supported by collecting data in different timeframes for *S. aor* from multiple sites. Sensitivity analysis across bin sizes and uncertainty quantification via Monte Carlo and bootstrap methods confirmed the robustness of the estimates, aligning with best practices in fisheries research (Froese et al., 2016).

CHAPTER 04 RESULTS

4 RESUTS

4.1 Observed output of aLBI R Package

To assess the stock status of *Sperata aor*, the collected length-frequency data were analyzed using the functions of the newly developed *aLBI* R package. The package's core functionalities were employed to generate the necessary parameters and plots for a comprehensive evaluation of the stock status.

4.1.1 Construction of Frequency Table

The initial step of the analysis involved the construction of a length-frequency table from the raw data using the *FrequencyTable* function. This function was utilized to create tables with various bin sizes, ranging from 1 cm to 5 cm, to examine how different class widths influenced the data distribution. The function also automatically calculated an optimum bin size (OBS) of 3 cm, which was determined using a statistically robust formula to minimize information loss. This 3 cm bin size was subsequently adopted as the base model for the study to ensure a reliable and scientifically sound foundation for all further analyses. The upper length class observed in the generated tables varied from 76 cm for the 1 cm bin size to 79 cm for the 5 cm bin size, highlighting the direct relationship between bin width and the structure of the resulting length distribution.

4.1.2 Frequency distribution plot

The total lengths (TL) of 1177 *S. aor* individuals, ranging from 9.5 to 75 cm were measured. Base model frequency table with a optimum bin size 3cm was used in FishPar function which produce a unimodal length frequency distribution plot (Figure 4.1) was produced by aLBI package. The observed frequencies represented by bars, revels the highest abundance of individuals within the length class of 27 cm (156 individuals) and followed by 30 cm (143 individuals), 21 cm (137 individuals), 24 cm, 18cm both with the same count (135 individuals), indicating these lengths are common among the sampled fish population.

The red curve, representing the modeled frequency, aligns closely with the observed data, highlighting the accuracy of the applied smoothing approach. This curve is generated using a loess (locally estimated scatterplot smoothing) regression, to fit a smooth line to the histogram, enhancing visual interpretation of the length-frequency distribution. The plot was found to be heavily skewed towards right side, indicating good recruitment but also potential growth overfishing.

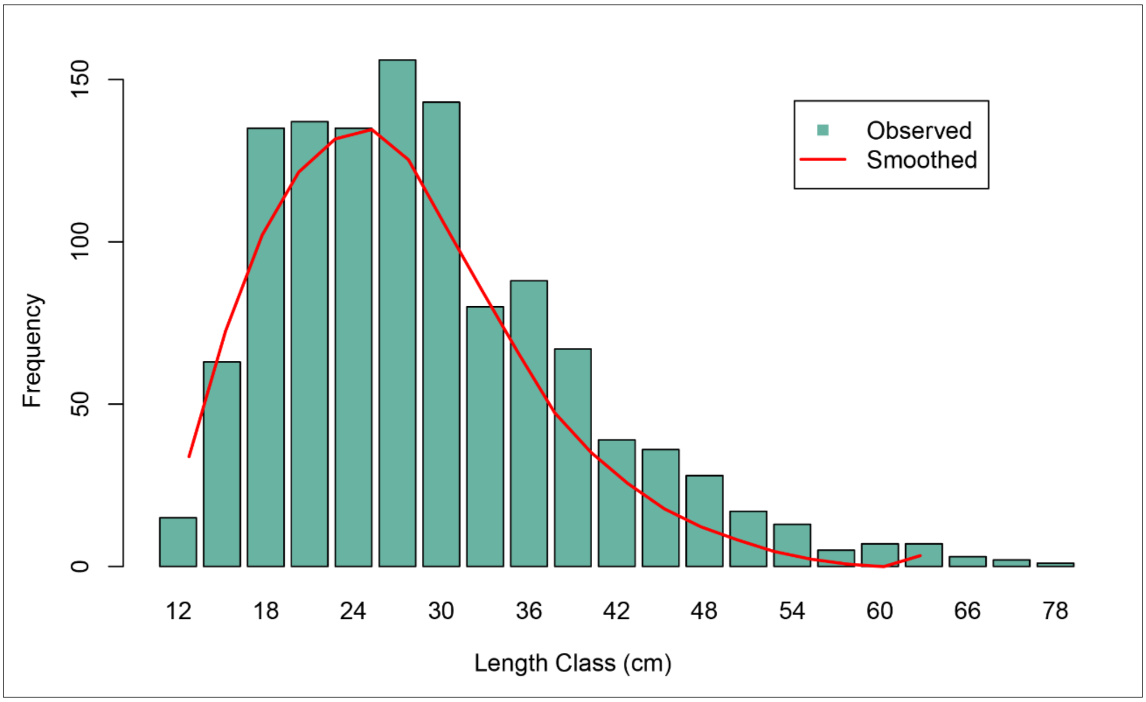

Figure 4.1: Plot showing the length frequency distribution of *S. aor*.

4.1.3 Estimating different length groups

The aLBI package produced a plot (Figure 4.2) for different length groups of *S. aor*. The length-based metrics provided a comprehensive understanding of the fish stock status. The plot depicts the mean maximum length (L_{max}) of 78.01 cm with a confidence interval (CI) 76.00 to 79.93 cm and asymptotic length (L_{∞}) of 82.12 cm with a CI of 80.00 to 84.13 cm, indicating that the species has the potential to attain considerable sizes under optimal conditions and minimal fishing pressure. The mean length at sexual maturity (L_{mat}) of 43.73 cm with a CI 40.71 to 46.78 cm. The mean optimum length (L_{opt}) is 47.03 cm with a CI 42.49 cm to 51.61 cm. On the contrary, the estimated range for L_{mat}±10% of L_{opt} is 42.33 (with a CI 38.24 to 46.45) to 51.74 cm (with a CI 46.74 to 56.77) (Table 4.1), represents the ideal range which produces maximum biological yield.

In Figure 4.2, the upper bounds are consistently close to the medians, indicating narrow confidence intervals. This results from the low variation in the resampled length data and the large sample size, both of which lead to higher precision in the bootstrap estimates.

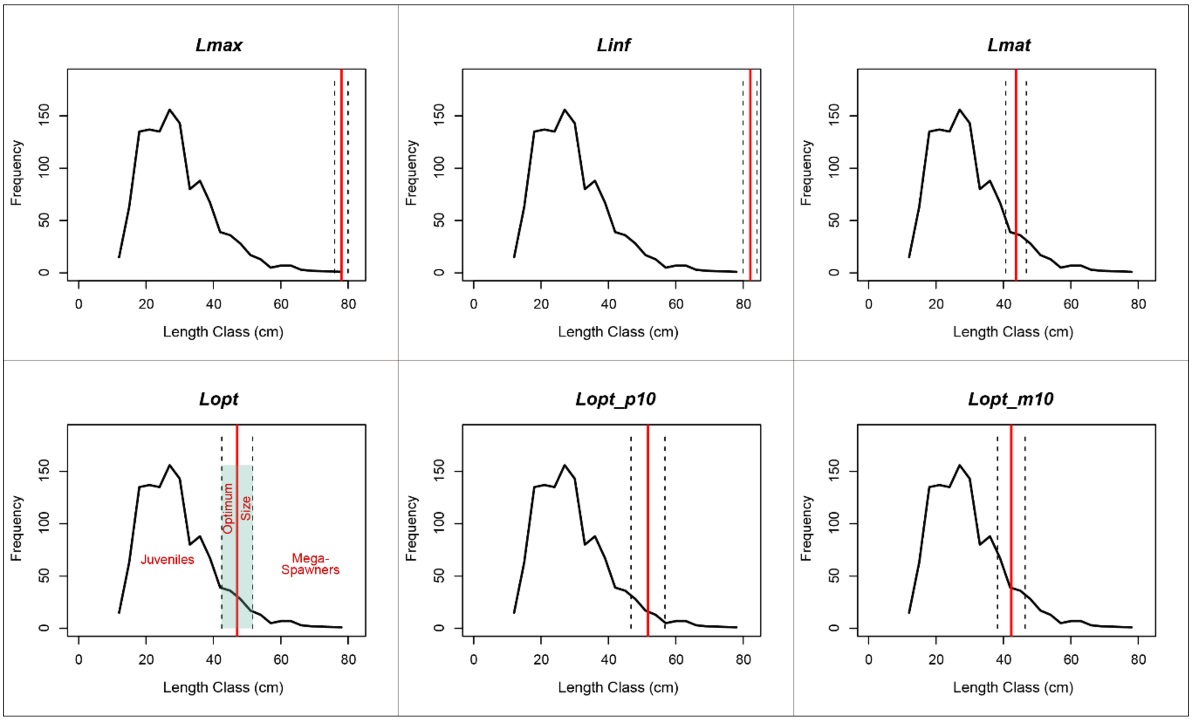

Figure 4.2: Plot showing the frequency line with different length parameters of Seperata aur. Solid red lines depict the mean estimates and dotted lines depicts the lower (LCI) and upper confidence intervals (UCI).

Estimated length-based indicators of *S. aor* and their respective confidence intervals are shown in the Table 4.1.

Table 4.1: Different length indicators of *S. aor* with 95% Confidence Intervals.

Indicators	Mean (cm)	LCI	UCI
L_∞	82.12	80.00	84.13
L_{max}	78.01	76.00	79.93
L_{mat}	43.73	40.71	46.78
L_{opt}	47.03	42.49	51.61
L_{opt_p10}	51.74	46.74	56.77
L_{opt_m10}	42.33	38.24	46.45

Note: LCI = Lower Confidence Interval; UCI = Upper Confidence Interval

4.2 Scenario produced with different bin size for length indicators

The boxplot (Figure 4.3) illustrates the consistency of the estimated length indicators (L_∞, L_{mat}, L_{max}, L_{opt}, L_{opt_m10}, and L_{opt_p10}) across different bin sizes (1 to 5, with the base model bin size of 3 cm). Though no statistical difference in the metrics when using different length bins, there are some differences in the values were observed. The estimated values of L_∞ and L_{max} exhibited moderate deviations from the base model, ranging from -2.53% to +1.27%, with the largest negative deviation at bin size 1cm (-2.53%) and positive deviation at bin size 5cm

(+1.27%). L_{mat} and L_{opt} showed deviations of -2.13% to +1.19% and -2.45% to +1.25%, respectively, with bin size 1cm showing the greatest reduction and bin size 5cm the largest increase relative base model bin size. Parameters L_{opt_m10} and L_{opt_p10} followed similar patterns, with deviations from -2.46% to +1.25%. Statistical analysis using One-Way ANOVA and Tukey's post-hoc test confirmed that the differences in estimated length indicators across different bin sizes were not significant ($p > 0.05$), suggesting low overall sensitivity.

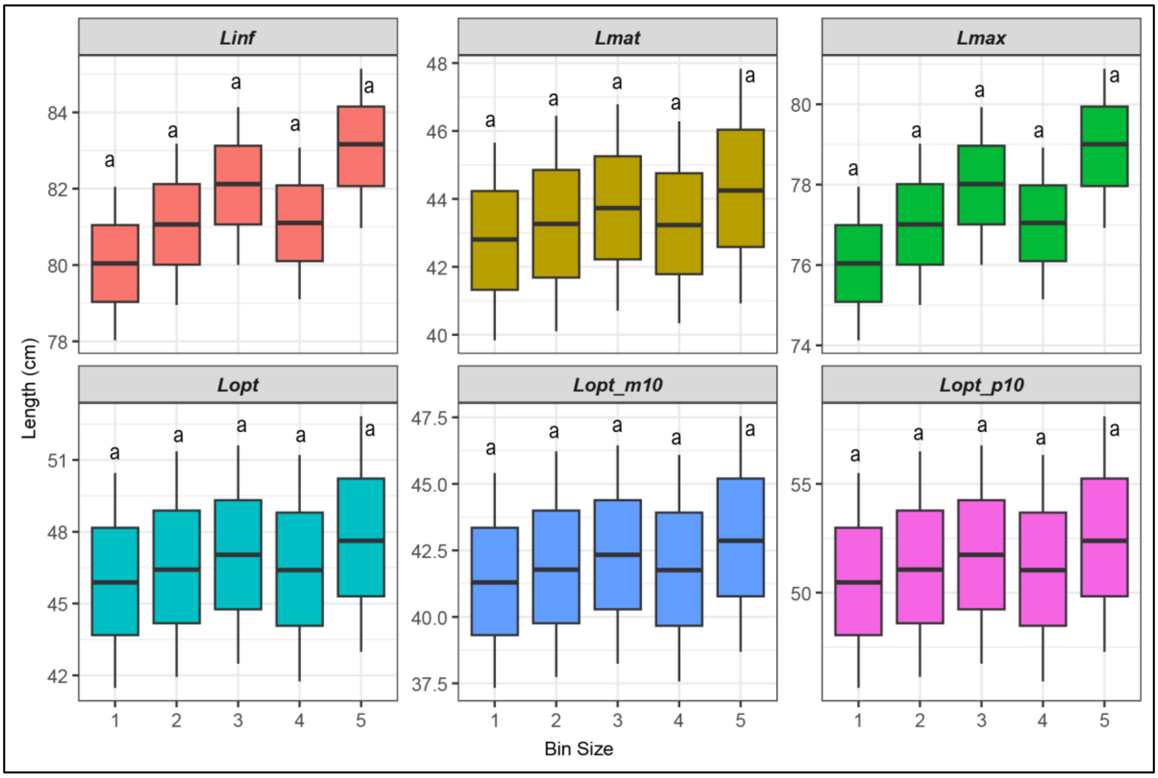

Figure 4.3: Distribution of length indicators with different bin size (base model bin size = 3 cm), represented with the mean, lower and upper confidence intervals. Shared letter (a) indicating no significant differences ($p > 0.05$) in the estimation of length indicators with different bin size based on one-way ANOVA and Tukey's HSD.

4.2.1 Length-Based Indicators (LBIs)

The simple set of indicators (P_{mat}. P_{opt}, P_{mega}) proposed by Froese (2004), were estimated for the stock of *Sperate aor* (Table 4.2) reveal significant deviations from the target reference values. The percentage of sexually mature fish (P_{mat}) was 2.04% (0.88–3.58%), far below the target of 100%, indicating heavy exploitation of immature fish and reduced reproductive potential. Similarly, the percentage of optimally sized fish (P_{opt}) was 2.38% (0.87–5.80%), is also below the 100% target, reflecting insufficient numbers of fish reaching their full growth potential. The proportion of mega-spawners (P_{mega}), crucial for genetic diversity and population resilience, was only 0.35% (0.08–1.16%), compared to the target of 20%, highlighting that this fishery has already removed major portion of its larger individuals. The combined indicator (P_{obj}) was 4.77% (1.84–10.53%), further underscoring the stock's unsustainable status.

Table 4.2: Estimated length-based indicators (LBIs) with 95% Confidence Intervals.

Indicators	Mean (%)	LCI	UCI
P_{mat}	2.04	0.88	3.58
P_{opt}	2.38	0.87	5.80
P_{mega}	0.35	0.08	1.16
P_{obj}	4.77	1.84	10.53

4.2.2 Scenario produced with different bin size for the aLBI indicators

The boxplot (Figure 4.4) illustrates the robustness of the method implemented in the aLBI package by producing consistent estimates of Froese sustainability indicators (P_{mat}, P_{opt}, P_{mega}) across varying bin sizes (1 to 5 cm, with a base model bin size of 3 cm). The propagated uncertainty from the length indicators (L_{mat}, L_{opt} and L_{max}) resulted in minor deviations from the base model, with P_{mat} ranging from -2.60% to +0.13%, P_{opt} from -3.25% to -0.4%, and P_{mega} from -1.07% to +1.53%. Statistical analysis using One-Way ANOVA and Tukey's post-hoc test confirmed that the differences in estimated length indicators across different bin sizes were not significant ($p > 0.05$), underscoring the method's stability and low sensitivity to bin size variation. These findings suggest that the aLBI package provides reliable estimates for fisheries management, though the low Pmat value may warrant further validation against species-specific maturity data.

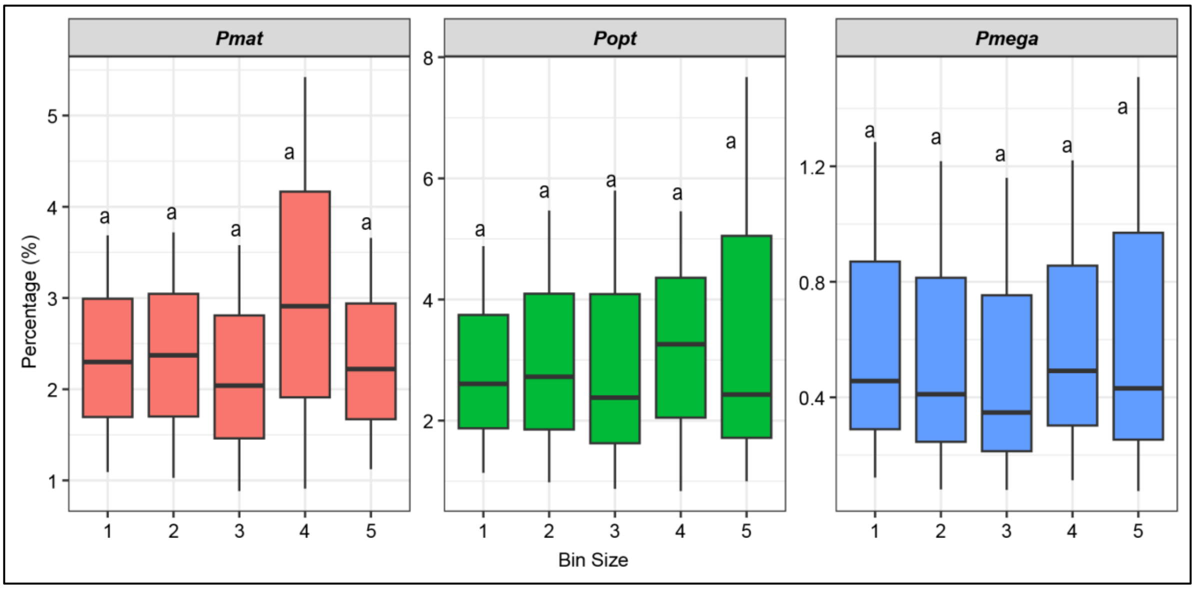

Figure 4.4: Distribution of LBIs with different bin size, represented with the mean, lower and upper confidence intervals. Shared letter (a) indicating no significant differences (*p > 0.05*) in the estimation of length indicators with different bin size based on one-way ANOVA and Tukey's HSD.

4.3 Observed stock status of *S. aor* by FishSS function

FishSS function facilitate the evaluation of stock status by quantifying the decision tree process outlined by Cope and Punt (2009). The sum of the all the three proportions (P_{mat}, P_{opt}, P_{mega}), termed as P_{obj}, provides comprehensive information to distinguish the selectivity patterns. The FishSS function estimates two key reference points: i) TSB40 (target spawning biomass at 40% of unfished levels) and ii) LSB25 (limit spawning biomass at 25% of unfished levels) based on input values including P_{obj}, P_{mat}, P_{opt}, and *LM_ratio*. Each P_{obj} is associated with these reference points, where the function automatically assesses whether the stock biomass is at or below the *TSB40* and *LSB25*. However, the findings (Table 4.4) of this study indicates that there is a probability that the stock biomass of *S. aor* is below both the target (*TSB40*) and the limit biomass level (*LSB25*) across all bin sizes.

Table 4.3: Estimated stock status of *the S. aor* by aLBI package.

Bin size (cm)	*LM*_ratio	P_{obj}	Probability (%) < TSB40	Probability (%) < LSB25
1	0.93	5.36	67	67
2	0.93	5.51	67	67
3	**0.93**	**4.77**	**67**	**67**
4	0.93	6.66	67	67
5	0.93	5.08	67	67

4.4 Sensitivity Analysis

The sensitivity analysis was conducted using nine distinct scenarios to evaluate the robustness of the aLBI R package. The Base Model (*BM*) was established with a length at infinity (L_∞) of 82.12 cm and a length at first maturity (L_{mat}) of 43.73 cm. To create the subsequent scenarios (S1-S8), the two key input parameters, L_∞ and L_{mat}, were systematically varied (Table 4.4). For scenarios S1 and S2, only the L_∞ value was adjusted, while for scenarios S3 and S4, only the Lmat value was changed. The remaining four scenarios (S5-S8) were designed to test combinations of these parameter variations. In each case, only one of the two parameters was altered at a time, with the other held constant at its base model value. This controlled approach allowed for a robust, single-parameter-at-a-time analysis, providing a clear understanding of how each input independently influences the package's estimations under realistic data-limited conditions.

Table 4.4 Input Parameters for Sensitivity Analysis.

Scenarios	Input Parameters	
	L_∞	L_{mat}
Base Model (BM)	82.12	43.73
S1	86.2*	-
S2	78.0*	-
S3	-	45.9*
S4	-	41.5*
S5	86.2	43.73
S6	78.0	43.73
S7	82.12	45.9
S8	82.12	41.5

** only used a single input for produce sensitivity scenario.*

4.4.1 Sensitivity Analysis for Length Parameters

The sensitivity analysis demonstrated that the aLBI R package's estimated length parameters are responsive to variations in the user-defined inputs, specifically L_∞, L_{mat}. A ±5% change in the input parameters resulted in a direct and proportional shift of approximately 4.5% to 5.0% in the estimated L_∞ and L_{mat} values (Figure 4.5). More critically, this input variation also led to significant changes in key indicators, with the median estimated L_{opt} and L_{opt_m10} shifting by up to 4.1% and 3.5%, respectively, while the median L_{opt_p10} varied by up to 5.2%. This quantitative breakdown confirms that while the model is robust, the reliability of its output for critical stock status indicators is highly contingent on the accuracy of the user-provided growth and maturity parameters.

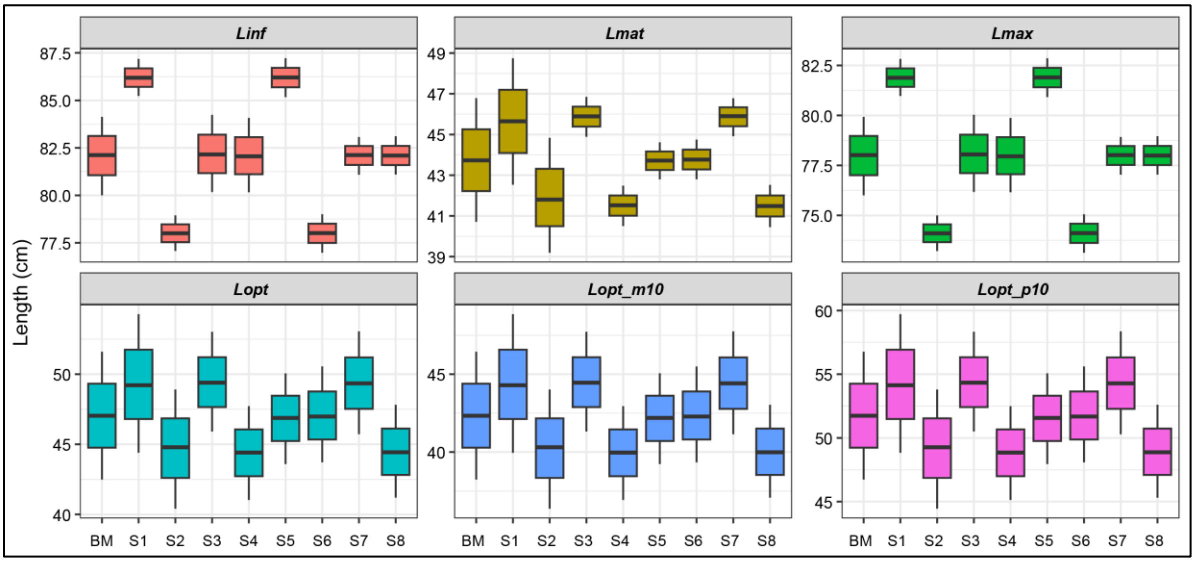

Figure 4.5: Boxplot Illustrating the sensitivity analysis of length parameters.

4.4.2 Sensitivity Analysis of Froese Length-based Indicators

The sensitivity analysis of the Froese Length-Based Indicators revealed that while all three are responsive to input uncertainty, the degree of impact varies significantly. The Pmat indicator showed a substantial quantitative response, with its median value, which was approximately 2.1% in the base model, shifting dramatically to as low as 1.2% and as high as 3.1% in different scenarios (Figure 4.6). Similarly, the median of Popt demonstrated a considerable change, with its base model value of 2.3% rising to nearly 3.5% in certain sensitive scenarios. In stark contrast, the Pmega indicator displayed extreme volatility, with its median dropping from 0.6% in the base model to as low as 0.3% and showing no consistent pattern, as evidenced by its wide interquartile ranges and overlapping distributions. This numerical evidence underscores that while P_{mat} and P_{opt} can be useful metrics for assessing a fishery's status, the high instability of Pmega makes it an unreliable indicator under conditions of input uncertainty, highlighting the critical importance of robust initial parameter estimation.

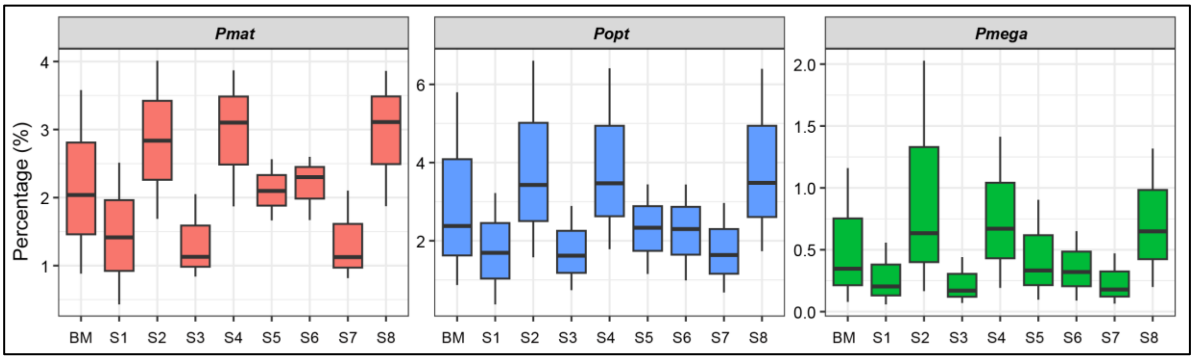

Figure 4.6 Boxplot Illustrating the sensitivity analysis of Froese LBIs.

CHAPTER 05 DISCUSSION

5 DISCUSSION

5.1 Development and application of the aLBI package

The aLBI package combines Froese's length-based indicators with the decision tree approach proposed by Cope and Punt (2009) to produce more robust estimates for the sustainable management of fisheries. In data-limited contexts, Froese's indicators are simple and easy to understand and facilitate rapid assessments and effective communication. However, this approach does not account for gear selectivity, which can influence the length composition of the catch. To address this limitation, Cope and Punt (2009) introduced a decision tree framework that integrates Froese's indicators with assumptions about selectivity to evaluate stock status relative to reference points, such as spawning biomass. By applying both methodological approaches in combination, aLBI offers assessments that are both accessible and scientifically rigorous, making it suitable for diverse fisheries management needs.

5.2 Development and application of the aLBI package

The aLBI package was developed to provide a practical tool for fish stock assessment using length-frequency data, particularly where conventional data sources are lacking. The package generates a whole framework for computing key length-based indicators like L_{max}, L_∞, L_{mat}, and L_{opt} as well as percentage indicators including P_{mat}, P_{opt}, and P_{mega}. These tools enable data-driven insights capable of guiding sustainable fisheries management. The ability of the program to visualize length frequency distributions and calculate confidence intervals helps managers and fisheries researchers to evaluate the health of any fish stock along with the uncertainties associate with the estimated parameters. Thus, this package will help decision-making for effective fisheries management by providing robust fisheries reference points.

5.3 User Accessibility and Interoperability

A central tenet of the "aLBI" package's design is to create a simple and intuitive interface supported by extensive, clear documentation. The package aims to significantly lower the barrier to entry for practitioners by providing straightforward functions, comprehensive user instructions, and practical, real-world examples. This approach ensures that a wider spectrum of users, including those without a background in statistical programming, can conduct high-quality stock assessments. As Wickham (2015) notes, user-friendly tools are essential for the widespread adoption of analytical methods in scientific fields. Furthermore, a foundational requirement for this package is its seamless integration and interoperability with other R packages and data sources. This capability allows users to perform comprehensive analyses that combine length-based indicators with other data types, a feature that is critical for holistic and multi-faceted stock assessments in modern fisheries management (Maunder & Punt, 2013). This interoperability improves the package's value and usability in the field, helping to support sustainable fisheries management in line with FAO's (2020) objectives.

5.4 Flexibility and Customization

The *aLBI* package is built with flexibility and customization as core principles to meet the diverse needs of different fisheries environments. This adaptability ensures that the program can be effectively applied in a wide variety of scenarios, from inland freshwater fisheries to coastal marine stocks. By providing options for modifying the assessment to meet specific demands, the package empowers users to tailor their analysis rather than being forced into a rigid, one-size-fits-all model. This strong and adaptable character allows fisheries scientists to analyze length-frequency data, compute biological parameters, and evaluate stock status with confidence, providing a comprehensive and versatile tool for both research and practical management.

By addressing these critical elements, this study seeks to fill a significant void in the current suite of instruments available for fish stock evaluation. The development of this R package will not only increase the scientific capacity of fisheries researchers but also directly contribute to sustainable management strategies that are vital for maintaining marine biodiversity and ensuring global food security

5.5 Performance evaluation of aLBI package

5.5.1 Frequency table and distribution curve

The aLBI package automatically calculates the optimal bin size for length-frequency data using the formula proposed by Wang et al. (2020). However, the FrequencyTable function of the package also allows users to specify a custom bin size and thus, ensure the package's flexibility, enabling the users to construct their frequency table according to their needs. For the *S. aor* stock status evaluation, the optimum bin size estimated by the package was 3.07 cm which was rounded to 3 cm to use as the base model bin size. With this bin size, the FishPar function of the package constructed the frequency distribution curve for the species which was skewed towards the smaller length classes. The dominance with smaller-length individuals indicates this fishery includes fish from the juvenile stage (Thorson et al., 2014b). Mia et al. (2024) and Bashar et al. (2021) reported a similar selectivity pattern for *Ailia coila* and *Eutropiichthys vacha* in this Kaptai Lake. According to the "The Protection and Conservation of Fish Act, 1950", catching fish with a length smaller than 23 cm is prohibited in this Lake (Rayhan et al., 2021). However, present and previous findings (Mia et al., 2024; Bashar et al., 2021) revealed the violation of this rules because of the absence of proper monitoring settings.

5.5.2 Different length indicators and their uncertainty

The length indicators such as L_{max}, L_{∞}, L_{mat}, and L_{opt} are important matrices for fisheries management. L_{max} serves as an indicator of the population structure and is closely associate with the fishing activities. The mean length at maturity (L_{mat}) underscores the critical threshold for reproductive capacity, while catching fish at the optimum length (L_{opt}) maximizes biological yield. These metrics emphasize the necessity of implementing size limits to protect both juveniles and mature fish. No prior studies have investigated these parameters for *S. aor* in Kaptai Lake.

Table 5.1: Estimated length parameters from different region of South Asia.

Regions	L_{∞}	L_{max}	L_{opt}	L_{mat}	Reference
Bangladesh	98.95*	94.00	56.65*	52.66*	Rahman, 2005
India	125.26*	119	70.80*	65.08*	Khan et al., 2019
India	192.42*	182.8	106.24*	95.68*	Day, 1878; Misra, 1959
India	128.42*	122.00	72.49*	66.55*	Job et al., 1955
India	166.44*	158.12*	92.63*	84.00	Saigal, 1964
India	108.72*	103.28*	61.92*	57.30	Ramakrishniah, 1992
Narora–Kanpur, India	96.50	90.10	54.43*	50.69*	Nazir & Khan, 2020
Varanasi, India	92.30	89.10	53.85*	50.19*	Nazir & Khan, 2020
Bhagalpur, India	89.10	85.40	51.74*	48.31*	Nazir & Khan, 2020
India	85.68*	81.40	49.44*	46.27*	Nazir & Khan, 2019
Pakistan	189.47*	180.00	104.70*	94.36*	Talwar & Jhingran, 1991; Tripathi, 1996
Bangladesh	82.12	78.01	47.03	43.73	Present study

* *Calculated by using motioned formula in the methodology from L_{max} or L_{mat}*

Previous studies conducted in various South Asian regions (Day, 1878; Misra, 1959; Talwar and Jhingran, 1991; Tripathi, 1996; Rahman, 2005; Sigal, 1964; Ramakrishniah, 1992; Khan et al., 2019; Job et al., 1955) have reported estimates for L_{max}, L_{∞}, L_{opt}, and L_{mat} that are significantly higher than the present study's findings (Table 5.1). While habitat quality and environmental conditions are critical factors influencing fish growth (Garcia et al., 2003; Alam et al., 2013), intense fishing pressure (especially high removal of juveniles) plays a major role in reducing the mean length of fish populations (Alam et al., 2021). This reduction in mean length contributes to declines in key length-based parameters and drives earlier maturation, as fish are increasingly exploited before they reach their full growth potential (Alam et al., 2021).

This study employed a bootstrapping resampling technique to address uncertainties in the estimates by calculating 95% confidence intervals for each parameter. The estimated parameters exhibited varying degrees of precision. For example, L_{mat} (40.71-46.78cm) and L_{opt} (42.49-51.6cm) showed relatively narrow confidence intervals, indicating higher precision, though others sources of uncertainty, such as those associated with growth parameters, are not accounted for. On the other hand, length-based indicators (LBIs) such as P_{mat} (0.88 to 3.58%), P_{opt} (0.87 to 5.80%), P_{mega} (0.08 to 1.16%) displayed narrow confidence intervals, reflecting lower uncertainty.

These uncertainties provide valuable information for fisheries management. While the estimated means of the LBIs clearly indicate overfishing and the depletion of mature, optimally sized, and large individuals, the confidence intervals - though relatively wide for P_{mat}, P_{opt} and P_{mega} do not overlap with their respective target thresholds (100% for P_{mat} and P_{opt}, and 20% for P_{mega}). Even under the most optimistic bounds, all estimates remain extremely below these targets, indicating depleted stock conditions. Meanwhile, the relatively narrow confidence intervals for Froese sustainability indicators *(LBIs)*enhance confidence in using these parameters to support size-based regulatory measures.

Different bin size analysis demonstrated a clear distribution of length parameters, showcasing central tendencies, interquartile ranges, and variability. The robustness of the modeling approach was assessed based on the consistency of estimates for L_∞, L_{mat}, L_{max}, L_{opt}, L_{opt_p10}, and L_{opt_m10} across different bin sizes. The differences among the estimated values for different parameters were remained within a ±2.5% deviation. Statistical analysis further confirmed that there were no significant differences ($p > 0.05$) of those estimates across different bin size. Thus, the performance of the package is not significantly affected by the choice of different bin size. These confirmations are critical for the reliability in length-based fish stock assessment particularly in a data-poor condition (Hordyk et al., 2015). These findings established the robustness of the package for the stock assessment of any commercially important fish species when only length-frequency data are available.

It is important to note that accurately estimating L_∞ from length-frequency data alone is inherently difficult and highly dependent on the quality and representativeness of the data. This becomes especially problematic when L_∞ is derived indirectly from L_{max} using empirical formula, as it introduces additional uncertainty. Therefore, for the improvement of the robustness of the estimates, further work could explore the way to account for this uncertainty, for example by inputting alternative L_∞ values or distributions into the functions and assessing the sensitivity of the indicators to these assumptions.

5.5.3 Froese sustainability indicators (LBIs) of *S. aor*

The findings of the aLBI package for the *S. aor* highlighted the critical insights into the current stock health status of the fishery. A significantly lower percentage of sexually mature fish (2.04%) indicates that majority of the individuals (approximately 98%) are removed from the population in their premature stage, indicating the growth overfishing status of the fishery which significantly reduces reproductive capacity and long-term sustainability of the stock. On the hand, only a minor percentage of the catch composition were reported as optimally sized (2.38%) and mega spawner (0.35%), indicating the recruitment over-fishing status of the fishery. The significant depletion of mega-spawners (P_{mega} < 20%) suggests heavy fishing pressure on larger, older individuals, threatening genetic diversity and resilience (Mace & Sissenwine, 1993; Mora et al., 2009; Barneche et al., 2018; Rudd & Thorson, 2018). Thus, management strategies should be developed for ensuring the immature individuals can grow and spawn at least once before they are caught and the mega spawners remain intake (Barrowman & Myers, 1996; Hilborn, 2007; Alam at el., 2022). These strategies will maintain the reproductive success of the population (Froese, 2004; Hilborn et al., 2020). These results align with earlier research showing that overfishing usually results in an overpopulation of young fish, which, under poor management, could cause recruitment overfishing (Beverton & Holt, 1957; Cushing, 1988). Furthermore, ensuring the genetic variety and long-term population health depends on keeping a balanced size distribution (Hixon et al., 2014).

5.5.4 Evaluation of stock status

The stock status probabilities for target spawning biomass (*TSB40*) and limit spawning biomass (*LSB25*) across various bin sizes are below their respective target reference points. Moreover, the ratio of length for sexually mature fish to the optimally-sized fish ($L_{mat}/L_{opt} > 0.9$) and the combined percentage of optimally-sized and the mega spawners ($P_{opt} + P_{mega} > 0$) indicate selectivity patterns favoring the capture of smaller and optimally-sized fish while avoiding the larger individuals. Since, there is no strategy for releasing the mega spawners from the catch, this finding also indicates the absence of larger *S. aor* individuals in this Lake. Previous studies (Mia et al., 2024; Bashar et al., 2021) also reported low percentage of mega spawners for *A. coila* and *E. vacha* in this Lake. Given the multi-gear and multi-species nature of *Kaptai lake* fisheries it can be concluded that major portion of mega spawners of different species have already been removed from this Lake by the existing selectivity pattern of the fisheries.

5.5.5 Sensitivity Test of aLBI R Package

The sensitivity analysis of the aLBI R package confirmed that its estimated parameters and indicators are directly influenced by changes in user-defined inputs, specifically L_∞, L_{mat}. The analysis revealed a two-tiered pattern of sensitivity.

First, the core length parameters, such as L_∞, L_{mat}, demonstrated a predictable and robust response. The boxplots for these parameters showed narrow distributions that precisely mirrored the controlled ±5% shifts in the input values. This confirms that the model is functioning as expected and that its base estimates are tightly linked to the provided data. However, the analysis also revealed that the reliability of the model's outputs for critical stock status indicators is highly dependent on the accuracy of these initial parameters.

Second, the Froese Length-Based Indicators (LBIs)— P_{mat}, P_{opt}, P_{mega}—exhibited varying degrees of sensitivity. Both Pmat and Popt showed a predictable, yet strong, response to input changes, with clear shifts in their median values and interquartile ranges. This highlights their direct link to the input parameters and suggests they can provide useful insights into stock status, provided the input data is reliable. In contrast, the Pmega indicator displayed significant instability, with its boxplots showing high variability and overlapping distributions across all scenarios. This extreme sensitivity makes Pmega an unreliable metric under conditions of input uncertainty, underscoring the critical need for precise parameter estimation for this particular indicator.

CHAPTER 06 CONCLUSION

6 CONCLUSION

The length-based indicators (LBIs) for fisheries sustainability proposed by Froese (2004), have been widely accepted and employed successfully in many regions globally. Building on the effectiveness of length-based indicators, this study developed an R package, 'aLBI', to provide more reliable and robust estimates of these indicators, and employed the package to evaluate the stock status of *S. aor* from Kaptai Lake, Chittagong, Bangladesh. In addition to the estimation of the length-based indicators, the package also introduced automated procedures to address the uncertainties associated with these estimates. The impacts of varying bin sizes were also examined in the evaluation of stock status of the *S. aor*, no significant differences were found when different bin sizes were applied. These indicates the effectiveness of the package compared to traditional LBI estimation methods in MS Excel. The outputs estimated by the aLBI package revealed that this fishery is suffering from both growth and recruitment overfishing and the current spawning biomass (SB) of the stock is below the reference points. To protect the juvenile and for the sustainability of the fishery, this study recommended a lower length limit of 47 cm which is the optimum length for this fishery.

6.1 Recommendations

Use size restrictions to stop early and immature fish from being captured, therefore enabling them to grow to maturity and assist the reproductive success of the population. Regulations and compliance monitoring help to enforce size restrictions, therefore guaranteeing young fish the chance to reach reproductive age.

Moreover, creating rules guarantees fish can attain their ideal weight before consumption. This approach will optimize the commercial value of the catch and assist to preserve a balanced population distribution. Aiming for bigger, more valuable individuals not only increases the natural survivability of the stock but also increases the profitability of fisheries.

Furthermore, creating no-take areas or seasonal closures helps to safeguard most reproducibly valuable species. Mega spawners are therefore rather important for preserving the genetic variety and general condition of the fish population. By helping to stabilize the population and guarantee enough spawners to maintain the stock over the long run, protecting these individuals helps to guarantee this.

Additionally, the aLBI tool allows one to monitor the fish stock constantly in order to evaluate changes in population dynamics and stock condition. By means of these assessments, changing management methods will help to guarantee sustainable development and adjust to changing conditions. Fast data produced by regular monitoring can guide adaptive management strategies and enable modifications in response to animal health or environmental conditions.

Apart from these, community-based management entails interacting with stakeholders like legislators, fishermen, and the society to encourage knowledge and support of sustainable fisheries management practices. Outreach campaigns and education can help to boost rule compliance and hence the efficacy of conservation measures. Including stakeholders in the management process means that several points of view are taken into account and that management techniques are practically feasible and socially acceptable.

Promote new studies to increase the capacity of the aLBI package. Among the developments might be new indicators, better visualization tools, and more general applications to many fish species and habitats. Thus, one more effective tool for fisheries management since continuous research helps address current constraints and increase the accuracy and usability of the package.

6.2 Limitations of the study

Despite the promising results, a key limitation of this study is the inherent sensitivity of the aLBI R package's outputs to the accuracy of its user-defined input parameters, a finding confirmed by the comprehensive sensitivity analysis. As demonstrated, even a small ±5% variation in the base model values for $L\infty$ and Lmat resulted in significant shifts in the estimated length-based indicators, with the Pmega indicator proving particularly volatile and unreliable under such conditions. This underscores a broader challenge for data-limited fisheries, where precise input parameters are often unavailable. Furthermore, the findings are based on a single species (*Sperata aor*) and a specific dataset, which may limit the generalizability of the results to other fish populations. Future research should therefore focus on testing the package across a wider range of species and datasets to more comprehensively evaluate its robustness and applicability.

Notwithstanding its advantages, the development and application of the aLBI package was limited in some respects. Data collecting proved challenging especially in severe weather, which regularly caused delays and partial data sets. This constraint highlights the need of robust and flexible data collecting methods adapted to environmental uncertainty.

Moreover, the development of the aLBI package required meticulous testing and validation to ensure dependability and correctness. Technical constraints and limited resources often hampered the development process, which stresses the importance of continuous investment in software development and user feedback mechanisms.

CHAPTER 07 REFERENCES

7 REFERENCES

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CHAPTER

08 APPENDIX

8 APPENDIX

8.1 Vignettes of the aLBI R package

8.1.1 Title

aLBI - A Simple R Package for Estimating Length-Based Indicators and Fish Stock Assessment from Length Frequency Data

8.1.2 Introduction

The aLBI package provides tools for estimating life history parameters, length-based indicators, and assessing fish stock using methods outlined by Cope and Punt (2009) and Froese (2004). These methods are particularly useful in data-limited situations and for providing simple indicators to address over-fishing and support sustainable fisheries management. This simple package facilitates the estimation of life history parameters of fish only from the length frequency data. Version 0.1.9 introduces the LWR function for length-weight relationship analysis and FreqMT for constructing frequency table for length data of multiple months. This version also updates FrequencyTable, FsihPar and FishSS with enhanced functionality. These includes the addition Monte Carlo, non-parametric bootstrapping, decision making based on selectivity and saving of outputs to the current working directory.

8.1.3 Function Overview

The aLBI package offers three primary functions: - **FrequencyTable**: Creates a frequency distribution table for fish length data using either a custom bin width or Wang's formula for automatic bin width calculation. The bin width is rounded to the nearest integer if calculated. - **FreqTM**: Creates frequency table for fish length of multiple months using either a custom bin width or Wang's formula as described in FrequencyTable function. - **FishPar**: This function calculates length-based indicators using Monte Carlo simulation for length parameters and non-parametric bootstrap for Froese indicators. - **FishSS**: This function assesses the stock status based on parameters calculated by the FishPar function. - **LWR**: Fits and visualizes length-weight relationships using linear regression.

8.1.4 Methods

The package includes functions to calculate various length-based indicators and visualize fish stock data. The approaches from Cope and Punt (2009) are used to establish reference points, while Froese (2004)'s indicators help to evaluate over-fishing status and selectivity. This vignette demonstrates how to use these functions with sample and provided data.

8.1.5 Installation

To install the aLBI package from CRAN or GitHub. To enjoy most recent updates, users are encouraged to install the aLBI from GitHub), follow these steps. Note that devtools is required to install packages from GitHub.

```
# You can install the package using the following commands in your R session:
# Install devtools if you haven't already
# install.packages("devtools")
# # Install dplyr if you haven't already
# install.packages("dplyr")
# # Install readxl if you haven't already
# install.packages("readxl")
# Install ggplot2 if you haven't already
# install.packages("ggplot2")
# Install openxlsx if you haven't already
# install.packages("openxlsx")
# Install rlang if you haven't already
# install.packages("rlang")
# install aLBI package from CRAN
# install.packages("aLBI")
# Install the most updated version of aLBI package from GitHub
# devtools::install_github("Ataher76/aLBI")
```

8.1.6 Package Management

Ensure the required packages are loaded in your R session. The following code checks for package availability and stops execution with a message if a package is missing.

```
# Check if required packages are installed and load them
# Check if required packages are installed and load them
required_packages <- c("aLBI", "readxl", "openxlsx", "dplyr", "devtools", "ggplot2",
"rlang")
for (pkg in required_packages) {
  if (!requireNamespace(pkg, quietly = TRUE)) {
    warning(paste("Package", pkg, "is required but not installed."))
  } else {
    library(pkg, character.only = TRUE)
  }
}

#> Loading required package: usethis
# Load the aLBI package
library(aLBI)
library(readxl)
library(dplyr)
library(ggplot2)
library(openxlsx)
library(rlang)
library(devtools)
```

8.1.7 Data Preparation for Functions

Prepare your collected data in a specific format (.xlsx or .csv) before using the functions. Here's an example of how to load and prepare your data using the readxl package:

8.1.8 Function (i): FrequencyTable

FrequencyTable function will generates the frequency table from the collected length data by calculated optimum bin size or user defined bin size. It also allows the user to provide Lmax from reliable sources such as **FishBase** or empirical studies. The function extract the length frequency data from the frequency table with the upper length_range and allow users to save the frequency data as an excel file for further use in the **FishPar** function.

8.1.8.1 Arguments

- **data**: A numeric vector or data frame containing fish length measurements. If a data frame is provided, the first numeric column is used.
- **bin_width**: Optional. A numeric value specifying the bin width for class intervals, if left NULL or if not provided, the bin width is automatically calculated using Wang et al. (2020) optimum bin size (OBS) formula formula.
- **Lmax**: Optional. The maximum observed length of fish. Required only if bin_width is NULL and Wang's formula is used. If not provided or left NULL then automatically calculate the frequency table with the observed maximum length of the provided data.
- **output_file**: Optional. Character string specifying the output Excel file name. Defaults to "FrequencyTable_Output.xlsx".

```
library(readxl)
# Load your length data from the system file
lenfreq_path <- system.file("exdata", "ExData.xlsx", package = "aLBI")
print(lenfreq_path)  # Check the generated path
#>
"C:/Users/User/AppData/Local/Temp/RtmpoHq14m/temp_libpath3fe442c399a/aLBI/exdata
/ExData.xlsx"

if (lenfreq_path == "") {
  stop("The required file ExData.xlsx is missing. Please check the inst/extdata directory.")
}

# load the lenght frequency data
lenght_data <- readxl::read_excel(lenfreq_path)
print(lenght_data)  # check the
#> # A tibble: 1,177 × 1
```

8.1.9 Function (ii): FreqTM

FreqTM Creates a frequency distribution table for fish length data across multiple months using a consistent length class structure. The bin width is determined by either a custom value or Wang's formula, applied uniformly across all months. The function dynamically detects and renames columns to 'Month' and 'Length' from the input dataframe. The maximum observed length is included as part of the last class, with the upper bound set to the smallest multiple of the bin width greater than or equal to the maximum length. Months can be converted to dates using a configurable day and year, with dates assigned sequentially in 'day.month.year' format (e.g., 15.01.26).

8.1.9.1 Arguments

- **data**: A data frame containing columns for months and lengths (names can vary, e.g., 'Month', 'Length', or any other names).
- **bin_width**: Optional. A numeric value specifying the bin width for class intervals, if left NULL or if not provided, the bin width is automatically calculated using Wang et al. (2020) optimum bin size (OBS) formula formula.
- **Lmax**: Optional. The maximum observed length of fish. Required only if bin_width is NULL and Wang's formula is used. If not provided or left NULL then automatically calculate the frequency table with the observed maximum length of the provided data.
- **date_config** A list with elements day (default 1) and year (default 2025) to set the day and year for converting month names to dates. The day must be between 1 and 31.
- **output_file**: Optional. Character string specifying the output Excel file name. Defaults to "FreqTM_Output.xlsx".

8.1.10 Function (iii): FishPar

The **FishPar** function estimates life history parameters (e.g., Lmax, Linf, Lmat, and Lopt) using Monte Carlo Simulation and Froese Sustainability Indicators using non-parametric bootstrapping method.

8.1.10.1 Arguments

- **data**: A data frame containing two columns: Length and Frequency.
- **resample**: An integer indicating the number of Monte Carlo samples or bootstrap resamples (default: 1000).
- **progress**: A logical value indicating whether to display a progress bar (default: FALSE).
- **Linf**: A numeric value for the asymptotic length (optional). If provided, overrides the default Lmax/0.95 calculation.
- **Linf_sd**: A numeric value for the standard deviation of random variation added to Linf (default: 0.5). Only used if Linf is provided.

- ❖ **Lmat**: A numeric value for the length at maturity (optional). If provided, overrides the default Monte Carlo estimation.
- ❖ **Lmat_sd**: A numeric value for the standard deviation of random variation added to Lmat (default: 0.5). Only used if Lmat is provided.

8.1.10.2 Explaination of Output

The function returns a list with the following components saved in three different sheet of excel file:

- **estimated_length_par**: Data frame of estimated length parameters with confidence intervals.
- **estimated_froese_par**: Data frame of estimated Froese indicators.
- **forese_ind_vs_target**: Data frame comparing Froese indicators with targets.
- **LM_ratio**: Length at first sexual maturity and optimum length ratio (LM_ratio = Lmat/Lopt).
- **Pobj**: Objective percentage combining Pmat, Popt, and Pmega.
- **Total_ind**: Number of total sampled individuals.

8.1.10.3 Produced Plot by FishPar Function

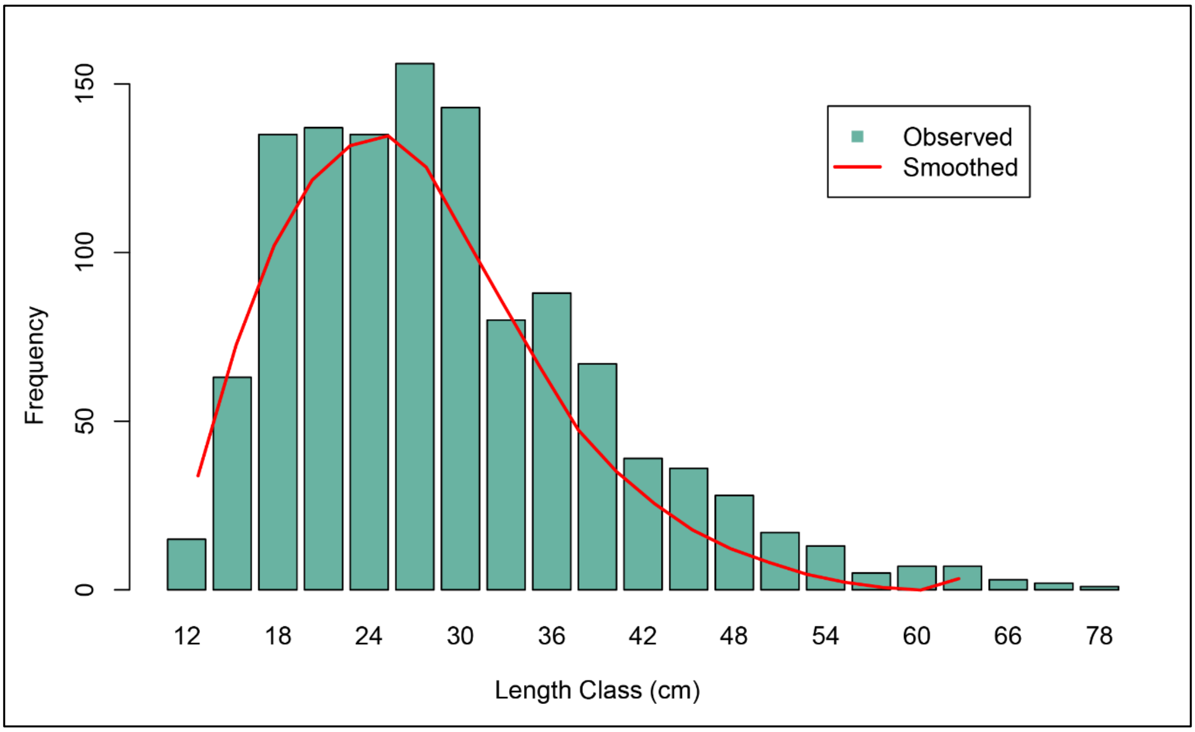

Figure 8.1: Length Frequency distribution plot.

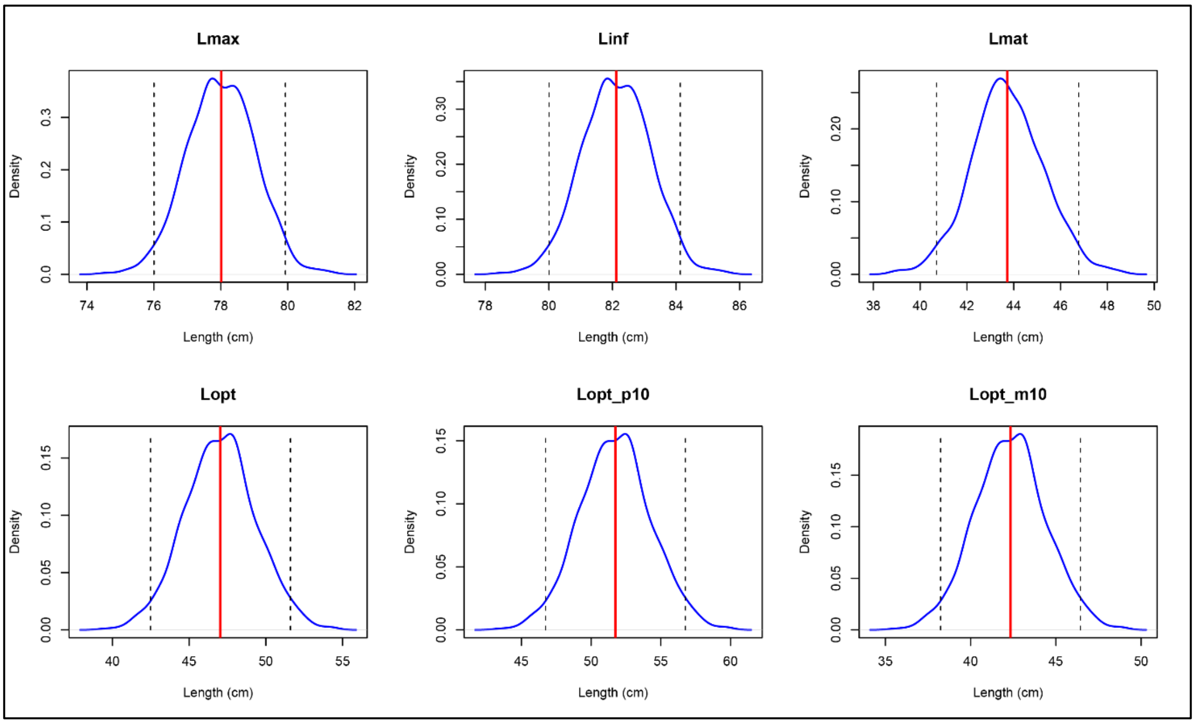

Figure 8.1: Density plot of Monte Carlo simulation for length parameters.

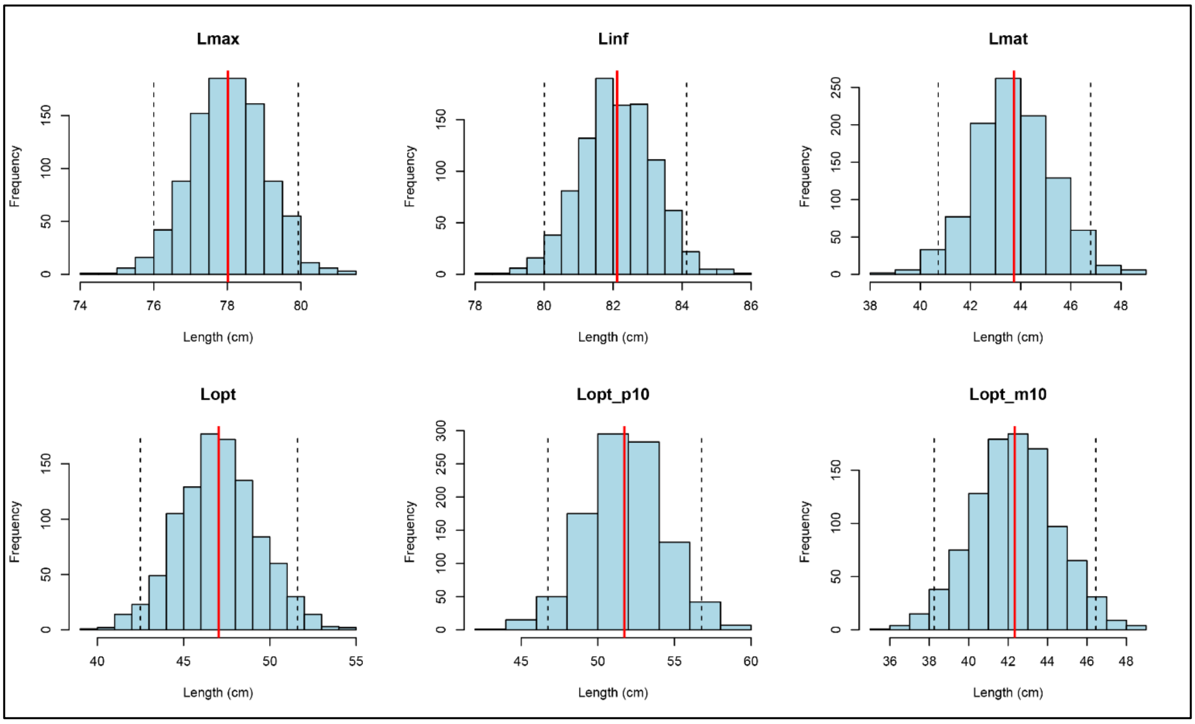

Figure 8.3: Histogram of Monte Carlo simulation for Length Parameters.

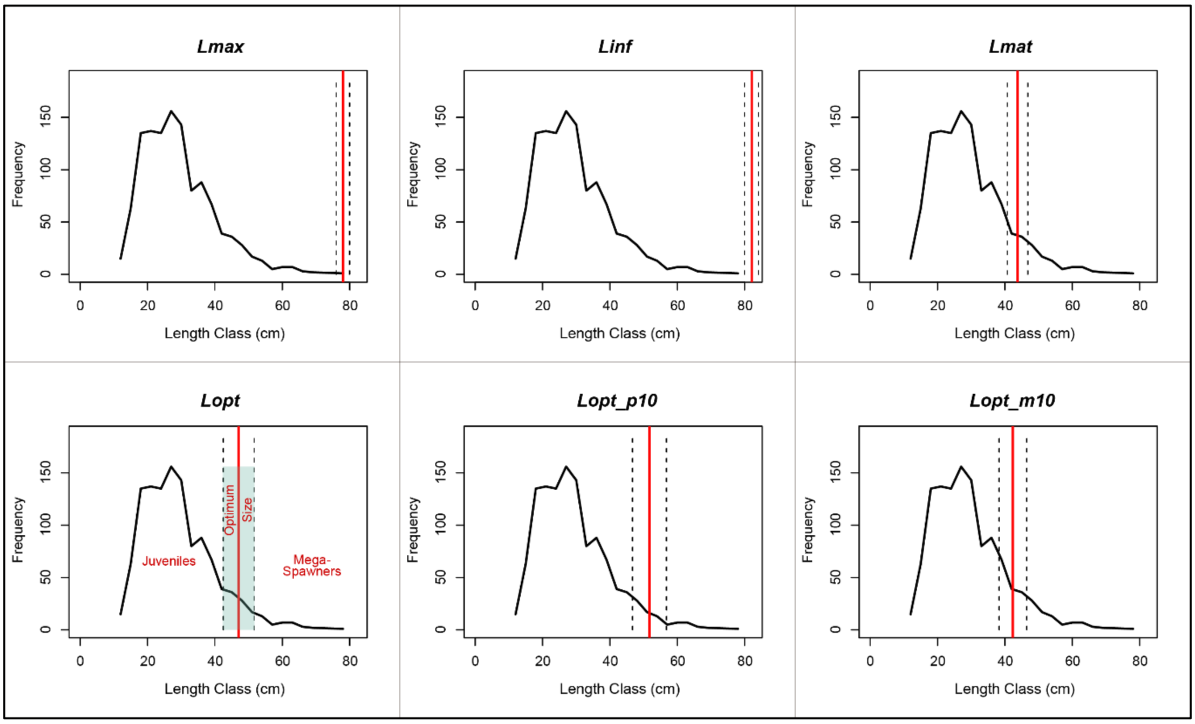

Figure 8.4: Main plot for length parameters with confidence intervals.

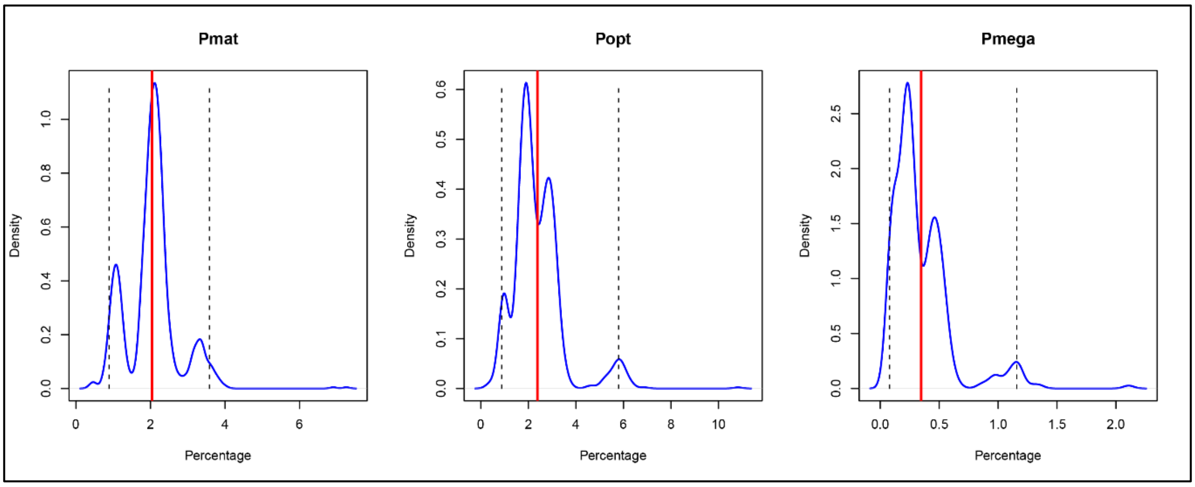

Figure 8.5: Density plot of non-parametric bootstrapping for Froese LBIs.

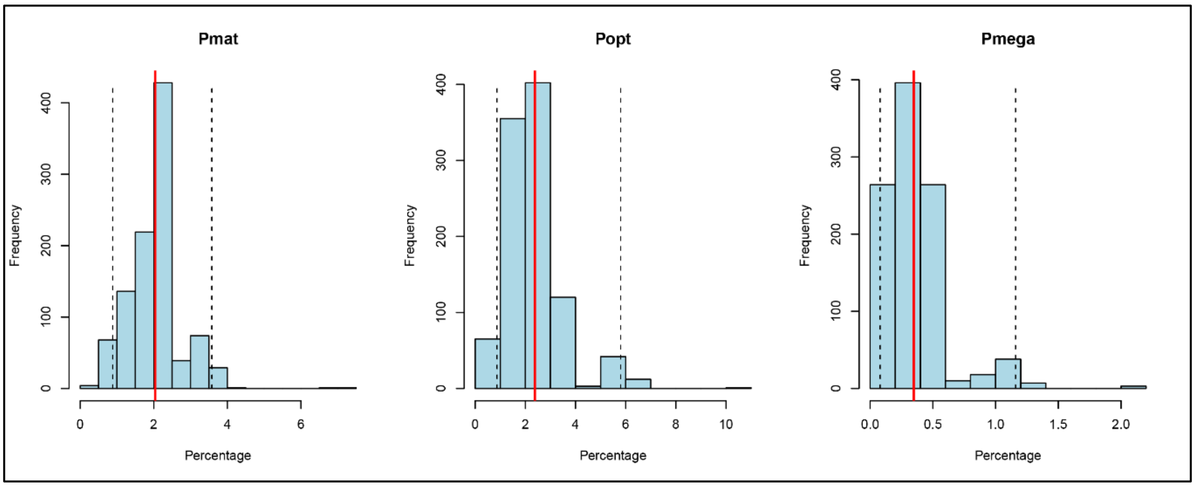

Figure 8.6: Histogram of non-parametric bootstrapping for Froese LBIs

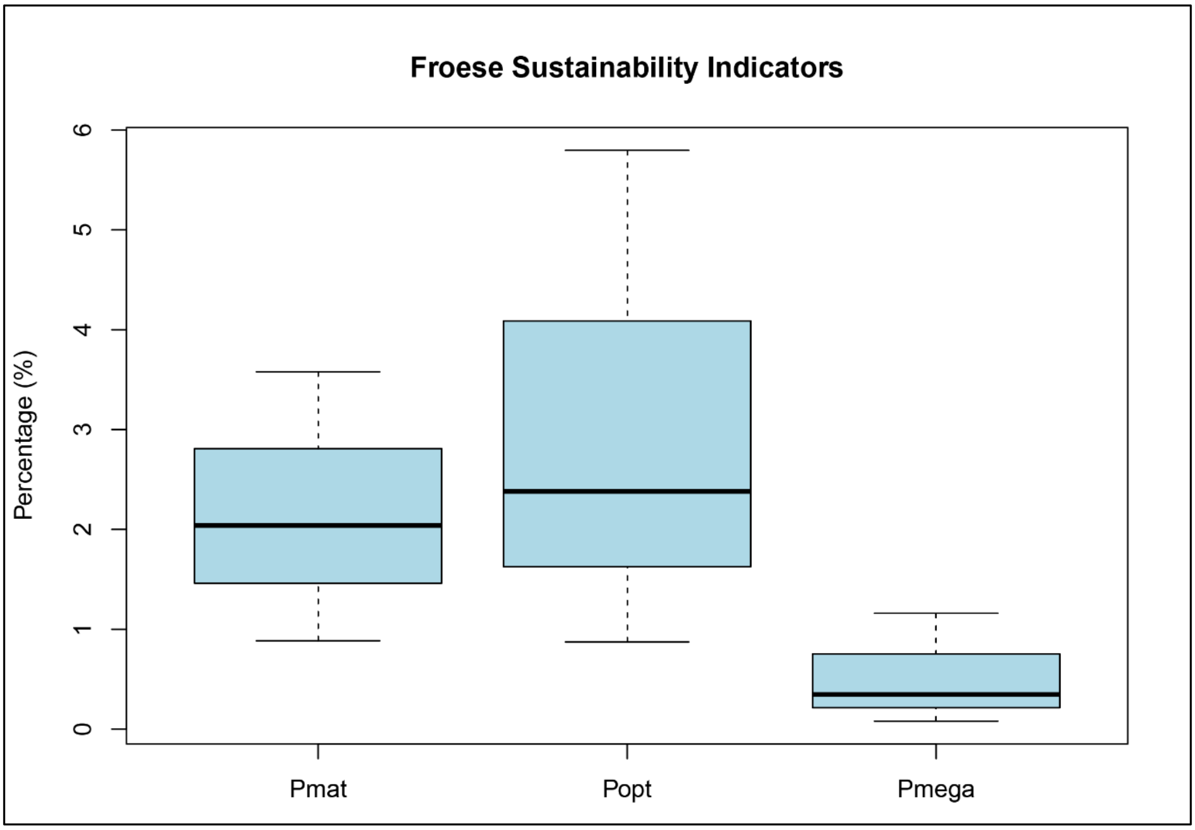

Figure: 8.7: Boxplot of Froese LBIs.

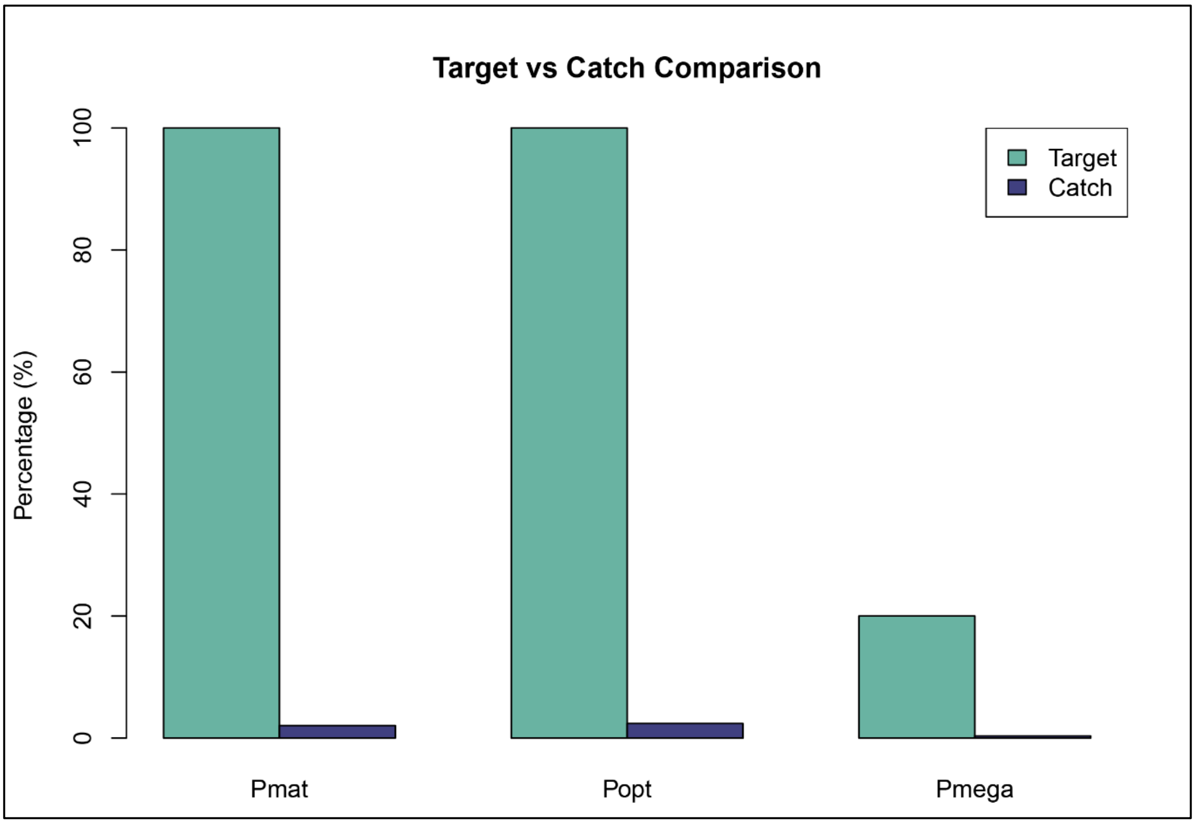

Figure 8.8: Bar plot of Froese target vs Catch in percentage

8.1.11 Function (iv): FishSS

The **FishSS** function evaluates stock status using criteria based on the estimated parameters by FishPar.

8.1.11.1 Arguments

data: A data frame containing the necessary columns for stock status calculation according to Cope and Punt (2009).

LM_ratio: A numeric value representing the length at first sexual maturity (Lmat) and Optimum length (Lopt) ratio.

Pmat: A numeric value representing the percentage of mature fish.

Popt: A numeric value representing the percentage of optimally sized fish.

Pmega: A numeric value representing the percentage of mega-spawner.

8.1.12 Function (iv): LWR

The **LWR** This function visualizes and models the relationship between length and weight (or any two continuous variables) using linear regression. It supports both standard and log-transformed models, producing a ggplot2-based plot with a fitted line, optional confidence interval shading, and annotations for the regression equation, R^2, and p-value. When save_output is TRUE, the plot and model summary are saved to the current working directory as a PDF and text file, respectively.

8.1.12.1 Arguments

- **data**: A data frame with two columns: the first for length, the second for weight.
- **log_transform**: Logical. Whether to apply a log-log transformation to the variables. Default is TRUE.
- **point_col**: Color of the data points. Default is black.
- **line_col**: Color of the regression line. Default is red.
- **shade_col**: Color for the confidence interval ribbon. Default is red.
- **point_size**: Size of the data points. Default is 2.
- **line_size**: Size of the regression line. Default is 1.
- **alpha**: Transparency for the confidence interval ribbon. Default is 0.2.
- **main**: Title of the plot. Default is "Length-Weight Relationship".
- **xlab**: Optional. Custom x-axis label. If , a label is generated based on log_transform.
- **ylab**: Optional. Custom y-axis label. If , a label is generated based on on log_transform.
- **save_output**: Logical. Whether to save the plot as a PDF and the model summary as a text file in the working directory. Default is TRUE.

8.1.13 Conclusion

The Fish Stock Assessment package provides a robust framework for estimating biological parameters and assessing fish stock status. By following the steps outlined in this vignette, you can effectively utilize this package for your fish stock assessment needs.

8.1.14 To read more detail

Readers are encouraged to read the article published in *Fisheries Research* of this aLBI R Package to understand the functionality and methodology.

Ali, A., Sarker, M. R., & Alam, M. S. (2025). Development of a simple R package (aLBI) for the estimation of stock status from the length frequency data. *Fisheries Research*, 288, 107467. https://doi.org/10.1016/j.fishres.2025.107467